

The Future of Growth and Human Labor Under Recursive AI Self-Improvement

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Abstract

I develop an endogenous-growth model in which a monopolistic developer supplies effective AI production services and uses effective AI research services to improve AI efficiency, creating recursive self-improvement. I characterize infinite-horizon equilibria with a finite AI-efficiency frontier. Long-run outcomes depend jointly on the substitutability between AI and labor and on the level of that frontier. When labor remains a production bottleneck, output per person and real wages grow at the exogenous rate of labor-augmenting technological progress, and labor retains a positive income share. With sufficient substitution and a sufficiently high frontier, the economy instead becomes AI-dominated: real-wage growth exceeds this exogenous rate, but output per person grows even faster. Labor's income share therefore converges to zero, and long-run growth and interest rates rise with the frontier. Numerical equilibrium paths show that moderate growth can persist for a long time before output accelerates and labor's income share falls sharply.

1 Introduction

Advances in artificial intelligence raise both hopes and fears. One hope is that increasingly capable AI could generate unprecedented economic prosperity. One fear is that workers who depend on labor income could lose economic power relative to the owners of capital and AI as labor's income share falls. This fear does not require real wages to decline: workers may receive a smaller share of a much larger economy even as their wages rise. Can unprecedented prosperity coexist with a vanishing labor income share?

I study this question by extending the Ramsey–Cass–Koopmans model. Competitive firms produce the final good with capital, effective labor, and effective AI production services. A monopolistic AI developer supplies those services and uses effective AI research services to improve AI efficiency, creating recursive self-improvement. AI efficiency improves endogenously but approaches a finite frontier. I ask how the substitutability between AI and labor shapes economic growth, real wages, interest rates, and the distribution of output.

The mechanism combines two feedback loops. A technological loop runs from AI efficiency to AI research: greater efficiency produces more effective research services from a given amount of research compute and thereby helps create still better AI. An economic loop runs from AI efficiency to output and back to research: better AI raises current production and expands the resources that can finance further improvements. This distinction follows [Davidson et al. \(2026\)](#). In my model,

household saving, capital accumulation, inference compute, research compute, and the developer's investment decision are determined jointly. Whether these feedbacks raise long-run growth depends on whether effective labor continues to constrain production.

I first characterize the labor-bottleneck regime. When AI and labor are complements, or when they are substitutes but attainable AI efficiency is not high enough, effective labor remains essential. Along the equilibria I construct, output per person and the real wage then grow at the exogenous rate of labor-augmenting technological progress. The net interest rate converges to its Ramsey value, and labor retains a positive share of output. Unit elasticity also belongs to this labor-supported class. Greater AI efficiency can change levels and transition dynamics without permanently raising growth above the exogenous rate.

The outcome changes when AI and labor are sufficiently substitutable and AI efficiency can reach a sufficiently high level. Gross substitution is therefore necessary but not sufficient for the economy to become AI-dominated. Above an explicit threshold, capital and effective AI production services can grow faster than effective labor. The growth rates of output per person and the real wage then exceed their labor-bottleneck benchmarks, as does the level of the interest rate. The real wage still grows, but output per person grows faster, so labor's income share converges to zero. Moreover, the long-run growth rate and the interest rate increase with the level of AI efficiency that the economy can attain.

The transition need not reveal this long-run regime quickly. I compute four equilibrium paths that share the same parameters, AI-efficiency frontier, and predetermined initial stocks, varying only the elasticity between AI and labor. In the illustrative AI-dominated case, output per person is still growing at only 1.062% and labor receives 60.5% of output after 500 model years. By year 1,500, growth has risen to 2.797% and labor's share has fallen to 14.5%. A long period of apparently moderate growth can therefore precede a sharp acceleration and a large redistribution of output. These model years are not calendar forecasts: the duration of the transition depends on an illustrative research calibration.

The paper's contribution is to characterize these growth and distributional outcomes jointly in a decentralized economy with endogenous saving, prices, compute allocation, AI investment, and a profit-maximizing developer. I give sufficient conditions for nearby infinite-horizon equilibrium trajectories, rather than only balanced-growth candidates, and report only numerical paths that satisfy the equilibrium-admission conditions. Because the representative household owns both capital and the AI developer, the model characterizes the functional distribution of output, not household inequality or political power. The existence results are local in the initial stocks and apply to each fixed finite AI-efficiency frontier; they do not establish equilibrium from arbitrary initial conditions or for an unbounded frontier. Section 2 relates the paper to the literature, Section 3 presents the model, Section 4 characterizes the equilibrium regimes, Section 5 presents the numerical paths, and Section 6 concludes.

2 Related literature

AI, growth, and automated research. The research technology builds on the endogenous- and semi-endogenous-growth tradition, in which nonrival knowledge and resources devoted to ideas determine long-run growth (Romer, 1990; Aghion and Howitt, 1992; Jones, 1995). Recent work asks how the conclusion changes when AI can automate production, research, or both. Aghion et al. (2019) show that automating idea production can accelerate growth but that tasks that remain difficult to automate can become binding bottlenecks. Trammell and Korinek (2023) distinguish the automation of ordinary production from the automation of R&D, while Jones (2025) emphasizes the share of research tasks that AI can perform, its productivity in those tasks, and the remaining bottlenecks. The closest feedback architecture is Davidson et al. (2026): more capable AI makes subsequent AI research more productive, while higher output expands the resources that can finance that research. I place these technological and economic feedbacks inside a decentralized Ramsey economy in which saving, inference compute, research compute, prices, and payments are equilibrium outcomes. This differs from quantitative scenario models such as Erdil et al. (2025), which combine AI development, automation, investment, and growth but abstract from the decentralized price system and market structure studied here.

Substitution, bottlenecks, and long-run growth. The production side connects this literature to the classic result that sufficient substitution toward accumulable inputs can prevent their marginal product from vanishing and produce an asymptotically linear technology (Pitchford, 1960; Jones and Manuelli, 1990; Rebelo, 1991). Duffy and Papageorgiou (2000) emphasize that a high elasticity is not by itself sufficient: the productivity of the accumulable input must also be high enough. In the AI literature, Nordhaus (2021) identifies substitutability between information and conventional inputs as central to an economic singularity. Jones and Tonetti (2026) show instead how a small number of weak links can delay growth acceleration even when automation advances, and Restrepo (2025) distinguishes work that remains a bottleneck from work that is no longer essential for growth. I obtain a related distinction without assuming that all work can be performed by AI. Gross substitution between AI services and labor is necessary but not sufficient for labor to cease being a bottleneck: attainable AI efficiency must also be high enough. I derive the threshold separating the labor-bottleneck and AI-dominated equilibria. The capability threshold in Gans (2025) concerns a different mechanism—whether AI directs research toward dense or exploratory regions of the knowledge space—rather than whether effective AI services can replace labor in final production.

Wages and the labor income share. The distributional results build on the task approach to automation. Acemoglu and Restrepo (2019) distinguish the displacement effect of automating existing tasks from the productivity and reinstatement effects that can raise labor demand. In CES models, the unit-elasticity threshold determines whether greater use of an input reduces the income share of the other input, a mechanism used by Karabarbounis and Neiman (2014) to study

the declining labor share. Recent work shows that greater automation can therefore raise wages while reducing labor’s income share (Autor and Kausik, 2026; Yango, 2026). In the long run, Ray and Mookherjee (2022) show that capital accumulation can drive labor’s share to zero while real wages rise indefinitely, and Restrepo (2025) obtains a vanishing labor share when compute can perform all bottleneck work. I do not claim either coexistence result as new. I show instead when it arises in the equilibrium with autonomous AI research: above the regime threshold, output per person and real wages both grow faster, but output grows faster than wages. The result yields an exact expression linking the output–wage growth wedge to the rate at which labor’s income share declines.

Interest rates. AI-driven growth can also affect equilibrium returns. Faster expected consumption growth raises long-term real rates through consumption smoothing in Chow et al. (2026), and the AI scenarios in Rachel (2025) and Melina and Villa (2025) can raise the natural rate through different saving and investment channels. The sign is not universal in richer models. I study a narrower object: the net return on physical capital in a closed-economy Ramsey equilibrium. The return remains tied to labor-augmenting technological progress when labor is the bottleneck, but rises with attainable AI efficiency in the AI-dominated regime. This is a regime-specific comparative static, not a forecast of market interest rates.

AI services and market structure. The distinction between raw compute and the effective services it produces is informed by Korinek and McKelvey (2026), who separately measure physical compute, inference, training and R&D, algorithms, and model capital. I retain separate compute flows for inference and research but summarize the technology that augments both in one AI-efficiency stock. The integrated developer is a tractable limiting case of the concentration forces studied by Korinek and Vipra (2024) and the AI market-power questions surveyed by Gans (2024). Evidence on model pricing, entry, and differentiation in Demirer et al. (2025) also cautions against treating current AI markets as a literal monopoly. I hold market structure fixed to isolate the dynamic mechanism. This accounting distinguishes the AI industry’s gross revenue from monopoly profit: inference and research expenditure are paid before the developer distributes the residual to its owner.

Contribution. The individual ingredients—research automation, production bottlenecks, declining labor shares, rising wages, and higher returns under faster growth—are established in the literature. My contribution is their joint equilibrium characterization in a model with recursive AI improvement and a profit-maximizing AI developer. I derive an explicit condition separating the labor-bottleneck and AI-dominated regimes, characterize the associated growth, wage, interest-rate, and income-share limits, and establish sufficient conditions for nearby infinite-horizon equilibrium trajectories rather than only balanced-growth candidates. The existence results are local in initial stocks and apply to each fixed finite AI-efficiency frontier; they do not claim global existence from arbitrary initial conditions or establish an equilibrium for an unbounded frontier.

3 Model

Time is continuous. The economy contains a representative household, competitive final-good firms, and an integrated AI developer. Labor-augmenting technology A grows exogenously. AI efficiency B improves through autonomous AI research: the only current input used to raise B is research compute. The model therefore describes an economy in which the replacement of human researchers at the technological frontier has already occurred.

3.1 Population and labor-augmenting technology

Population grows at the exogenous rate $n \geq 0$:

$$\dot{N} = nN, \quad N(0) = N_0 > 0. \quad (1)$$

Each person supplies one unit of homogeneous labor inelastically. The aggregate labor endowment is therefore N . Because autonomous research uses no human labor, all labor is employed in final production and equilibrium will impose $L = N$.

The variable A measures the level of labor-augmenting technology. One unit of labor supplies A efficiency units, so effective labor is AL . Labor-augmenting technology evolves according to

$$\dot{A} = \gamma A, \quad A(0) = A_0 > 0, \quad \gamma \geq 0. \quad (2)$$

3.2 Household

The representative household owns physical capital and the AI developer. Let Π denote the net distribution the household receives from the AI developer. The household takes the paths of the net interest rate r , the wage w and Π as given. For a predetermined capital stock $K_0 > 0$, it chooses aggregate consumption C to solve

$$\max_{\{C_t\}_{t \geq 0}} \int_0^\infty e^{-\rho t} N_t \log(C_t/N_t) dt, \quad \rho > n, \quad (3)$$

subject to

$$\dot{K} = rK + wN + \Pi - C. \quad (4)$$

The variable r is the net real interest rate: depreciation has already been deducted from the rental return to capital. The household's Euler condition is

$$\frac{\dot{C}}{C} = n + r - \rho. \quad (5)$$

Its transversality condition is

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0. \quad (6)$$

3.3 Final-good production

Competitive firms produce the final good from physical capital K , effective labor AL , and effective AI production services X . Let $0 < \alpha < 1$ be the physical-capital elasticity, $\delta \geq 0$ the depreciation rate, and $\sigma > 0$ the elasticity of substitution between effective labor and effective AI production services. The CES weights satisfy $\omega_L, \omega_X \in (0, 1)$ and $\omega_L + \omega_X = 1$. Production is

$$Y = K^\alpha Z^{1-\alpha}, \quad (7)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}. \quad (8)$$

Jobs comprise multiple tasks, and AI may substitute for human input in some tasks while complementing it in others (Autor, 2013; Acemoglu and Restrepo, 2019). I abstract from differences across workers, occupations, and tasks by modeling homogeneous labor and aggregate effective AI production services. The elasticity of substitution governs the relationship between these aggregate inputs, rather than the allocation of individual tasks to humans or AI. This simplification focuses the analysis on aggregate wages and labor's income share, leaving differences in exposure and earnings across workers outside the model.

Let p_X denote the price of one unit of effective AI production services. Taking the input prices $r + \delta, w$, and p_X as given, a final-good firm solves

$$\max_{K, L, X \geq 0} \{ K^\alpha Z(AL, X)^{1-\alpha} - (r + \delta)K - wL - p_X X \}. \quad (9)$$

Define s_X as the elasticity of Z with respect to X :

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}. \quad (10)$$

The interior first-order conditions are

$$r = \alpha \frac{Y}{K} - \delta, \quad w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad p_X = (1 - \alpha) s_X \frac{Y}{X}. \quad (11)$$

Constant returns and these conditions imply

$$Y = (r + \delta)K + wL + p_X X. \quad (12)$$

Solving the final-good firm's AI condition for its price yields the inverse demand schedule $p_X(X)$. The inverse absolute elasticity is

$$e_X \equiv -\frac{\partial \ln p_X(X)}{\partial \ln X} = \frac{1 - s_X}{\sigma} + \alpha s_X. \quad (13)$$

This demand schedule is an input to the developer's problem.

3.3.1 Effective AI production services and labor

The elasticity σ governs two different margins. The first is whether more effective AI production services raise or lower labor's marginal product. The second is whether labor receives a larger or smaller fraction of output. These margins have different thresholds. Define production labor's share of output as

$$\ell^Y \equiv \frac{wL}{Y} = (1 - \alpha)(1 - s_X). \quad (14)$$

Proposition 1 (Effective AI production services, wages, labor's income share, and conditional labor demand; [Autor and Kausik, 2026](#); [Yango, 2026](#)). *Consider an increase in X . To evaluate its effects on w and ℓ^Y , hold K and L fixed; to evaluate its effect on conditional labor demand L , hold K and w fixed. If $\sigma < 1$, w , ℓ^Y , and L all rise with X . If $1 < \sigma < 1/\alpha$, w and L rise while ℓ^Y falls. If $\sigma > 1/\alpha$, all three fall.*

Proof. The proposition follows from the final-good firm's first-order conditions:

$$\left. \frac{\partial \ln w}{\partial \ln X} \right|_{K,L} = \left(\frac{1}{\sigma} - \alpha \right) s_X, \quad (15)$$

$$\left. \frac{\partial \ln \ell^Y}{\partial \ln X} \right|_{K,L} = - \left(1 - \frac{1}{\sigma} \right) s_X. \quad (16)$$

$$\left. \frac{\partial \ln L}{\partial \ln X} \right|_{K,w} = \frac{s_X(\sigma^{-1} - \alpha)}{\alpha(1 - s_X) + s_X/\sigma}. \quad (17)$$

□

Proposition 1 describes conditional responses to an increase in effective AI production services X , not the general-equilibrium effects of greater AI efficiency. The latter depend on how the developer adjusts service supply and how capital and prices respond.

3.4 The AI developer

3.4.1 Effective AI production and research services

The model distinguishes two uses of compute. The variables U and M are flows of computing resources allocated to inference and research, respectively. In a concrete application, their empirical counterparts would be compute expenditure per unit of time. I choose to measure compute in units such that one unit of compute costs one unit of the final good.

The variable B measures AI efficiency. It is a nonrival scalar index of how effectively computing resources produce AI services. A higher B allows a given amount of compute to produce more effective services; equivalently, B is an effective-compute multiplier. The computer-science literature calls closely related improvements *algorithmic efficiency* or *compute-augmenting algorithmic progress* ([Hernandez and Brown, 2020](#); [Erdil and Besiroglu, 2022](#)). In AI measurement work, inference and R&D/training output are instead quality-adjusted for algorithmic progress ([Korinek](#)

and McKelvey, 2026). I use the broader term *AI efficiency* because the same B augments compute used in both activities. This distinction between compute and the AI technology that augments it is also consistent with Sastry et al. (2024). Inference compute produces effective AI production services according to

$$X = BU. \quad (18)$$

Equation (18) distinguishes the raw computing input U from the resulting flow $X = BU$ of effective AI production services. Research compute analogously produces the flow BM of effective AI research services. Thus U and M are rival resource inputs, while BU and BM are the corresponding effective service flows. In the terminology of Korinek and McKelvey (2026), their closest empirical counterparts are quality-adjusted inference output and quality-adjusted R&D/training output. The research use also corresponds to the AI-research feedback studied by Davidson et al. (2026).

Figure 1 summarizes the production block.

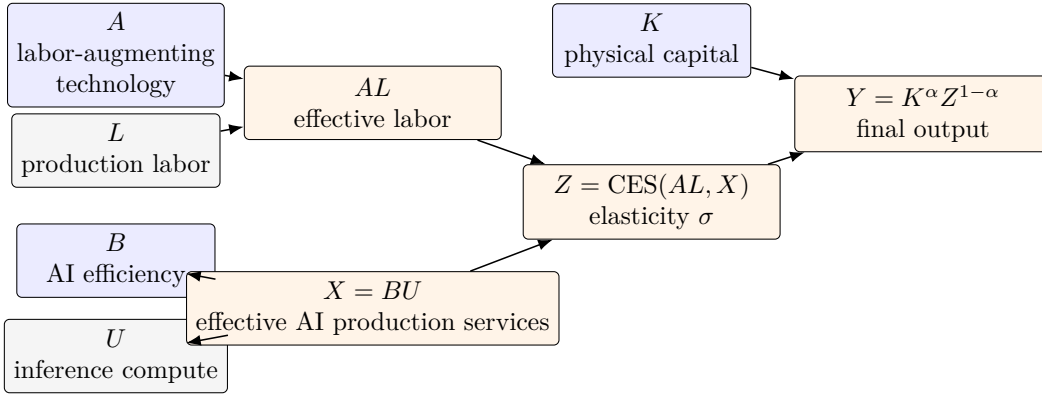


Figure 1: Production of final output and effective AI production services. AI efficiency B augments inference compute U to produce $X = BU$, the flow of effective AI production services. These services combine with effective labor AL to produce Z , which combines with capital K to produce Y .

3.4.2 Autonomous improvement in AI efficiency

Given initial AI efficiency $B_0 > 0$ and an efficiency frontier $\bar{B} \in (B_0, \infty]$, AI efficiency evolves according to

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad \psi(B) = 1 - \frac{B}{\bar{B}}, \quad \chi > 0, \quad 0 < \eta < 1. \quad (19)$$

Figure 2 summarizes the AI-efficiency improvement mechanism.

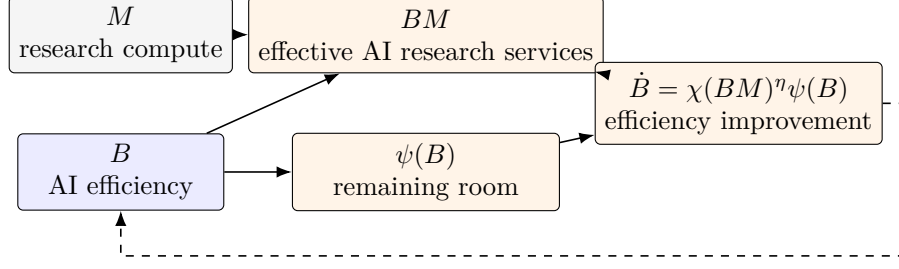


Figure 2: Autonomous improvement in AI efficiency. AI efficiency B augments research compute M to produce BM , the flow of effective AI research services. These services and the remaining room for improvement $\psi(B)$ determine $\dot{B}$, which increases B .

The parameter χ shifts research productivity. The parameter η governs returns to current research scale. The same AI-efficiency index B augments both inference and research compute. This common-productivity assumption is parsimonious but substantive. Because BM enters the law of motion for AI efficiency, a higher B makes each unit of research compute M more productive and creates recursive improvement. The term $\psi(B)$ reduces the remaining room for improvement as AI efficiency approaches $\bar{B}$. The uncapped specification is recovered at $\bar{B} = \infty$.

The efficiency frontier $\bar{B}$ allows me to establish and compute an equilibrium for each finite frontier and then ask whether these equilibria have a meaningful limit as $\bar{B} \rightarrow \infty$. Without the frontier, recursive improvement may instead produce a finite-time singularity, as discussed below. AI efficiency may approach the frontier asymptotically but cannot cross it in finite time.

3.4.3 Supply of effective AI production services

An integrated AI developer has exclusive ownership of B . It conducts research and sells effective AI production services. The developer sells X and purchases the $U = X/B$ units of inference compute needed to supply those services. Because X does not enter the law of motion for AI efficiency defined by Equation (19) directly, its within-date choice can be solved conditional on B, K , and L . Define optimized operating profit before research expenditure as

$$\pi(B) = \max_{X \geq 0} \left\{ p_X(X)X - \frac{X}{B} \right\}. \quad (20)$$

The notation $\pi(B)$ leaves its dependence on current K, A , and L implicit. The first-order condition equates marginal revenue to the compute cost of one additional service unit:

$$p_X(1 - e_X) = \frac{1}{B}. \quad (21)$$

For $K, A, L, B > 0$, condition (21) selects the unique global maximizer $X^* > 0$. Appendix A.2 gives the proof.

The monopoly price is therefore

$$p_X = \frac{1/B}{1 - e_X}. \quad (22)$$

The envelope theorem gives

$$\pi_B = \frac{X}{B^2} = \frac{U}{B}. \quad (23)$$

At a fixed X , a marginal increase in AI efficiency saves U/B units of the consumption good.

AI efficiency is nonrival in use. Once B has been developed, using it to augment one unit of inference or research compute does not reduce its ability to augment other units. This does not make effective AI services costless: BU and BM require the rival and costly compute inputs U and M . Nonrivalry also does not rule out excludability. An owner can restrict access to model weights, software, or AI services.

To see the resulting appropriability problem, suppose that competitive service providers had free access to B . Free entry would imply

$$p_X = \frac{1}{B}, \quad p_X X - \frac{X}{B} = 0.$$

The service market would generate no operating rent. A developer that paid for M would improve a common AI-efficiency index B that competing providers could use without compensating it. The developer would bear the research cost but could not appropriate the resulting benefits. Exclusive ownership allows the developer to capture part of the value created by a higher B and makes the return to research appropriable, as in the standard endogenous-growth treatment of nonrival ideas (Romer, 1990).

That being said, monopoly is an institutional assumption, not a technological implication of nonrivalry. Licensing, oligopoly, public funding, or open-source provision with research subsidies could also support investment in B . Each alternative would require an additional institution and would change prices, profits, and the distribution of rents. Recent work studies these alternatives in more detail (Korinek and Vipra, 2024; Gans, 2024; Demirer et al., 2025).

3.4.4 The dynamic problem

The AI developer uses M for research and owns B and any resulting improvements in AI efficiency. Given the solution to problem 20, the monopolist chooses how much research compute M to rent to increase AI efficiency B and solves the following problem:

$$\Pi = \max_{\{M_t\}_{t \geq 0}} \int_0^\infty \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt, \quad (24)$$

subject to (19) and B_0 given.

Profit can be negative at a date because the model imposes no separate cash-flow constraint. A negative Π represents an equity injection from the household that owns the developer.

Let q be the current-value shadow price of AI efficiency, measured in units of the current final good per unit of B . An interior research choice satisfies

$$q\chi\eta B(BM)^{\eta-1}\psi(B) = 1, \quad (25)$$

Equation (25) equates the marginal benefit of research compute to its unit cost. Holding current B fixed, an additional unit of M raises the rate of improvement in AI efficiency by $\chi\eta B(BM)^{\eta-1}\psi(B)$. Multiplication by q values the resulting efficiency gain, accounting for future profit gains and further improvements in AI efficiency.

The current-value costate equation is

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (26)$$

Rearranging gives the return equation

$$rq = \frac{U}{B} + q\eta \frac{\dot{B}}{B} + q\dot{B} \frac{\psi'(B)}{\psi(B)} + \dot{q}. \quad (27)$$

The first term on the right-hand side of Equation (26) is the inference-compute saving from greater AI efficiency. The second is the gain in research productivity. With a finite frontier, the third term is negative because a higher B leaves less room for further improvement. The final term is the capital gain on AI efficiency. Together these returns must equal the market return rq .

The developer's transversality condition is

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (28)$$

It rules out a path that leaves positive discounted value in the AI-efficiency state.

I maintain the following restriction.

Assumption 1 (Large-scale research curvature).

$$0 < \eta < \alpha < 1. \quad (29)$$

Appendix A.4 gives the supporting result, economic interpretation, and proof.

3.5 Equilibrium

I can now combine the conditions of the three agents. Capital-market clearing identifies household capital with the capital rented by final-good firms. Labor-market clearing gives $L = N$. AI-service market clearing identifies the developer's sales with the final sector's purchases. Combining the household budget, final-good zero profit, developer profit, and market clearing yields the aggregate resource constraint

$$C + \dot{K} + \delta K + U + M = Y. \quad (30)$$

Definition 1 (Equilibrium). *For exogenous initial values $A_0, N_0, K_0, B_0 > 0$, an equilibrium is an infinite-horizon trajectory*

$$\{K_t, B_t, C_t, q_t, L_t, U_t, M_t, X_t, Z_t, Y_t, r_t, w_t, p_{X,t}, s_{X,t}, e_{X,t}, \Pi_t\}_{t \geq 0} \quad (31)$$

such that for every date, the following conditions hold:

The exogenous paths and predetermined initial conditions are

$$A_t = A_0 e^{\gamma t}, \quad (32a)$$

$$N_t = N_0 e^{nt}, \quad (32b)$$

The intratemporal allocation and prices satisfy

$$L = N, \quad (33a)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad (33b)$$

$$Y = K^\alpha Z^{1-\alpha}, \quad (33c)$$

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}, \quad (33d)$$

$$e_X = \frac{1 - s_X}{\sigma} + \alpha s_X, \quad (33e)$$

$$r = \alpha \frac{Y}{K} - \delta, \quad (33f)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (33g)$$

$$p_X = (1 - \alpha) s_X \frac{Y}{X}, \quad (33h)$$

$$\frac{1}{B} = p_X (1 - e_X), \quad (33i)$$

$$X = BU, \quad (33j)$$

$$1 = q\chi\eta B(BM)^{\eta-1}\psi(B), \quad (33k)$$

$$\Pi = p_X X - U - M. \quad (33l)$$

The four dynamic equations are

$$\dot{K} = Y - C - U - M - \delta K, \quad (34a)$$

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad (34b)$$

$$\frac{\dot{C}}{C} = n + r - \rho, \quad (34c)$$

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (34d)$$

The two transversality conditions are

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0, \quad (35a)$$

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (35b)$$

Remark 2 (The economy without AI). As a separate boundary case, set $\omega_X = 0$ and $\omega_L = 1$, so that $Z = AL$. Effective AI production services have no productive use, and the developer has no revenue with which to reward improvements in AI efficiency. Thus $X = U = M = \Pi = 0$; B remains constant and does not affect allocations. With $L = N$, the model reduces to the Ramsey–Cass–Koopmans economy with logarithmic utility and exogenous labor-augmenting technological progress:

$$\begin{aligned} Y &= K^\alpha (AN)^{1-\alpha}, & \dot{K} &= Y - C - \delta K, \\ r &= \alpha Y/K - \delta, & \dot{C}/C &= n + r - \rho. \end{aligned}$$

The household transversality condition remains unchanged. The elasticity σ and the AI research parameters are irrelevant in this reduced economy; the interior AI price and costate conditions must not be imposed at $\omega_X = 0$. In particular, $wN/Y = 1 - \alpha$ and $(r + \delta)K/Y = \alpha$.

For a finite AI-efficiency frontier, Proposition 8 in Appendix A.5 provides a sufficient test for the developer’s global optimality.

The system contains no separate household budget or final-good zero-profit equation because these are redundant after market clearing. The consolidated income decomposition follows from the equations in the definition:

$$1 = \frac{(r + \delta)K}{Y} + \frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y}. \quad (36)$$

It records how final output is divided among gross capital income, labor income, developer profit, inference compute, and research compute. It is an accounting consequence of equilibrium, not an additional equilibrium condition.

For the analysis of equilibrium under complementarity, I use

$$0 < \sigma < 1, \quad \rho > n, \quad 0 < \eta < \alpha < 1, \quad 2\eta \leq 1.$$

The restriction on σ selects the complementarity case; it does not exclude the other substitution regimes from the model. The condition $\rho > n$ makes discounting dominate population growth in the household’s objective, whose effective discount rate is $\rho - n$.

The restriction $\eta < \alpha$ limits the feedback from research expenditure to AI efficiency and then to operating profit. Without a finite efficiency frontier, it ensures that, on a fixed finite horizon and taking K, A, L , and r as given, a bound on the operating profits generated by research grows less than proportionally with total research expenditure, while its cost grows linearly. Arbitrarily large expenditure therefore cannot be optimal. Appendix A.4 establishes this result. A finite frontier already bounds AI efficiency, so $\eta < \alpha$ is not needed for finite-horizon existence of the developer’s optimum in that case. I retain it as a sufficient restriction on research expenditure that also applies when the frontier is removed.

The condition $2\eta \leq 1$ serves a different purpose: it makes the research term $(BM)^\eta$ jointly

concave in B and M . Doubling both AI efficiency and research compute increases this term by at most a factor of two. For the finite-frontier technology, Appendix A.5 uses this concavity after a change of state variable. Together with the operating-profit curvature condition stated there, it makes the first-order and transversality conditions sufficient for global optimality of a feasible developer policy with a finite objective. It is not, by itself, a concavity condition for the full AI-efficiency law in the original state variable.

When $\alpha \leq 1/2$, the restriction $\eta < \alpha$ already implies $2\eta < 1$. I state both inequalities to distinguish control of research expenditure from the concavity argument. These restrictions do not by themselves establish the existence of an equilibrium trajectory; the remaining conditions on the frontier, initial stocks, and agent optimality must also be checked.

4 Equilibrium existence with a finite AI-efficiency frontier

I now give sufficient conditions for an infinite-horizon equilibrium in each substitution regime. Throughout this section, $0 < B_0 < \bar{B} < \infty$, $\chi > 0$, $n + \gamma > 0$, $\rho > n$, $0 < \eta < \alpha < 1$, and $2\eta \leq 1$; the remaining parameters satisfy Section 3. These results do not concern the uncapped economy. They are local in initial stocks, not in time: each constructs a trajectory defined for every $t \geq 0$, including both transversality conditions and agent optimality. The initial stocks need not lie on a balanced-growth path.

A finite AI-efficiency frontier does not necessarily make long-run growth of output per person equal to γ . It bounds AI efficiency, but not the compute used to supply effective AI production services. If AI and labor are substitutes, capital and compute can sustain growth faster than effective labor even when AI efficiency converges to a finite limit.

For any positive variable V , write $g_V = \dot{V}/V$. Define the output–capital ratio required for growth at the effective-labor rate as

$$\mathcal{R} = \frac{\rho + \gamma + \delta}{\alpha}. \quad (37)$$

Indeed, if K and C grow at $n + \gamma$, the Euler equation requires $r = \rho + \gamma$. For $\sigma \neq 1$, define

$$z_A(\bar{B}; \sigma) = \left[(1 - \alpha)^2 \bar{B} \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha}, \quad (38)$$

$$\bar{B}_c(\sigma) = \frac{\mathcal{R}^{\alpha/(1-\alpha)}}{(1 - \alpha)^2 \omega_X^{\sigma/(\sigma-1)}}. \quad (39)$$

The first expression is the limiting Y/K implied by the static equations when $s_X \rightarrow 1$ and $B \rightarrow \bar{B}$; the second solves $z_A(\bar{B}_c; \sigma) = \mathcal{R}$. Appendix A.6 derives these expressions and proves the following propositions. I use $\bar{B}_c(\sigma)$ as an economic regime threshold only when $\sigma > 1$. For $\sigma < 1$, it appears only as a sufficient condition for the particular local finite-frontier construction below.

4.1 Complementarity: $\sigma < 1$

Proposition 3 (Equilibrium trajectories with complementary inputs). *Maintain the common conditions of this section. If $0 < \sigma < 1$ and $\bar{B} > \bar{B}_c(\sigma)$, there exist constants $\bar{k}, \bar{d} > 0$ and a neighborhood $\mathcal{U}_L$ of $(\bar{k}, \bar{d}, 0)$ such that every positive initial configuration satisfying*

$$\left(\frac{K_0}{A_0 N_0}, (\bar{B} - B_0) A_0 N_0, \frac{1}{A_0 N_0} \right) \in \mathcal{U}_L \quad (40)$$

admits an interior infinite-horizon equilibrium. The initial values C_0 and q_0 are uniquely selected among nearby trajectories converging to this terminal regime. Along the equilibrium,

$$B \rightarrow \bar{B}, \quad g_Y - n \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \frac{wL}{Y} \rightarrow (1 - \alpha)(1 - \bar{s}_X) > 0, \quad (41)$$

where $\bar{s}_X \in (0, 1)$ is the limiting CES share.

With complementarity, both effective labor and effective AI production services remain relevant in this terminal regime. Effective AI production services expand through compute even as $g_B \rightarrow 0$, and output per person eventually grows at γ . The frontier condition ensures that AI efficiency is high enough to support this regime. If it fails, the proposition does not establish nonexistence of equilibrium; it only excludes the positive-ratio terminal construction used in its proof.

4.2 Unit elasticity: $\sigma = 1$

At unit elasticity I use the continuous CES limit $Z = (AL)^{\omega_L} X^{\omega_X}$, so $s_X = \omega_X$. Define $\beta = (1 - \alpha)\omega_X$. The monopoly condition gives $U = \beta^2 Y$.

Proposition 4 (Equilibrium trajectories at unit elasticity). *Maintain the common conditions of this section and let $\sigma = 1$. For every finite $\bar{B} > 0$, there is an interior infinite-horizon equilibrium for initial configurations satisfying (40) in a sufficiently small neighborhood, with*

$$\bar{k} = \left[(\beta^2 \bar{B})^\beta \mathcal{R}^{\beta-1} \right]^{1/(1-\alpha-\beta)}. \quad (42)$$

The positive constant $\bar{d}$ is constructed in the proof. The nearby convergent choice of C_0, q_0 is unique, and

$$B \rightarrow \bar{B}, \quad g_Y - n \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \frac{wL}{Y} = (1 - \alpha)\omega_L > 0. \quad (43)$$

The denominator in (42) is positive because $1 - \alpha - \beta = (1 - \alpha)\omega_L > 0$. No additional lower or upper bound on the finite frontier is needed for this local construction. The result describes transitions toward balanced growth, not an assumption that the economy starts on a balanced-growth path.

4.3 Gross substitution: $\sigma > 1$

Proposition 5 (Equilibrium trajectories with substitute inputs). *Maintain the common conditions of this section and let $\sigma > 1$.*

If $\bar{B} < \bar{B}_c(\sigma)$, initial configurations satisfying (40) in a sufficiently small neighborhood admit an interior infinite-horizon equilibrium with the limits in (41), for a limiting share $\bar{s}_X \in (0, 1)$.

If $\bar{B} > \bar{B}_c(\sigma)$, let $\varphi = (\sigma - 1)/\sigma$ and define

$$\bar{r} = \alpha z_A(\bar{B}; \sigma) - \delta, \quad \bar{g} = n + \bar{r} - \rho > n + \gamma. \quad (44)$$

There exist a constant $\bar{d} > 0$ and a neighborhood $\mathcal{U}_A$ of $(0, \bar{d}, 0)$ such that every positive initial configuration satisfying

$$\left[\left(\frac{A_0 N_0}{K_0} \right)^\varphi, (\bar{B} - B_0) K_0, \frac{1}{A_0 N_0} \right] \in \mathcal{U}_A \quad (45)$$

admits an interior infinite-horizon equilibrium with

$$B \rightarrow \bar{B}, \quad g_Y - n \rightarrow \bar{r} - \rho > \gamma, \quad \frac{wL}{Y} \rightarrow 0, \quad (46)$$

$$g_w \rightarrow \gamma + \frac{\bar{r} - \rho - \gamma}{\sigma} > \gamma. \quad (47)$$

In both cases the nearby convergent choice of C_0, q_0 is unique. The equality $\bar{B} = \bar{B}_c(\sigma)$ is not covered.

Above the threshold, a fixed AI-efficiency level allows output, capital, and inference compute to grow faster than effective labor. Production becomes asymptotically linear in capital, as in an AK model. Growth is finite for every fixed frontier, but need not equal γ . Real wages rise even as labor's income share vanishes. Below the threshold, labor retains a positive income share and long-run growth per person equals γ .

Proposition 6 (The output–wage growth wedge). *Along an AI-dominated equilibrium from Proposition 5, output per person grows asymptotically faster than the real wage:*

$$\lim_{t \rightarrow \infty} [(g_Y - n) - g_w] = \left(1 - \frac{1}{\sigma}\right) (\bar{r} - \rho - \gamma) > 0. \quad (48)$$

Because $L = N$, the same wedge is the asymptotic rate at which labor's income share falls:

$$\lim_{t \rightarrow \infty} \frac{d}{dt} \log \left(\frac{wL}{Y} \right) = - \left(1 - \frac{1}{\sigma}\right) (\bar{r} - \rho - \gamma) < 0. \quad (49)$$

Proof. Subtract the wage-growth limit in (47) from the per-person output-growth limit in (46). Moreover, $d \log(wL/Y)/dt = g_w + n - g_Y$ because $L = N$. $\square$

The result is not that workers must become poorer. Their real wage grows, but total output per person grows faster. Labor therefore receives a vanishing share of an expanding economy. The wedge is zero in the positive-labor-share regimes, where both growth rates converge to γ . Thus $\sigma > 1$ alone is insufficient: the frontier must also place the economy in the AI-dominated regime.

4.4 Deriving long-run growth and the effect of the frontier

The constructed trajectories have convergent growth rates and a positive limiting consumption–output ratio. Write $\bar{r} = \lim_{t \rightarrow \infty} r_t$. Consumption and output therefore have the same limiting growth rate. The household Euler equation then gives

$$\lim_{t \rightarrow \infty} \frac{d \log(Y_t/N_t)}{dt} = \lim_{t \rightarrow \infty} (g_C - n) = \bar{r} - \rho. \quad (50)$$

In the positive labor-share regimes, $Y/(AN)$ converges to a positive constant and its growth rate converges to zero. Consequently,

$$g_Y \rightarrow n + \gamma, \quad g_Y - n \rightarrow \gamma, \quad \bar{r} = \rho + \gamma. \quad (51)$$

These are limits of the constructed transitions, not restrictions imposed on their growth rates at every date.

In the AI-dominated regime, $s_X \rightarrow 1$ and $e_X \rightarrow \alpha$. The CES function and the monopoly condition imply

$$\frac{Z}{X} \rightarrow \omega_X^{\sigma/(\sigma-1)}, \quad \frac{U}{Y} \rightarrow (1 - \alpha)^2, \quad \frac{X}{Y} \rightarrow \bar{B}(1 - \alpha)^2. \quad (52)$$

Production gives the identity $Y/K = (Z/Y)^{(1-\alpha)/\alpha}$. Substitution of these limits therefore yields

$$\frac{Y}{K} \rightarrow z_A(\bar{B}; \sigma), \quad \bar{r} = \alpha z_A(\bar{B}; \sigma) - \delta, \quad \lim(g_Y - n) = \alpha z_A(\bar{B}; \sigma) - \delta - \rho. \quad (53)$$

The last expression exceeds γ exactly when $\bar{B} > \bar{B}_c(\sigma)$. A finite AI-efficiency level can thus support sustained growth above γ : inference compute expands with output, and effective labor ceases to constrain production asymptotically.

Holding $\sigma > 1$ and all other parameters fixed, the constructed families give the following comparative statics across finite frontiers:

$$\bar{r}(\bar{B}; \sigma) = \begin{cases} \rho + \gamma, & 0 < \bar{B} < \bar{B}_c(\sigma), \\ \alpha z_A(\bar{B}; \sigma) - \delta, & \bar{B} > \bar{B}_c(\sigma). \end{cases} \quad (54)$$

Below the threshold, a larger frontier does not change the limiting interest or growth rate. Above it,

$$\frac{\partial \bar{r}}{\partial \bar{B}} = \frac{1 - \alpha}{\bar{B}} z_A(\bar{B}; \sigma) > 0, \quad \lim_{\bar{B} \rightarrow \infty} \bar{r}(\bar{B}; \sigma) = \infty. \quad (55)$$

The limiting growth rate per person has the same derivative because it equals $\bar{r} - \rho$. The two interest-rate formulas approach $\rho + \gamma$ at the threshold, although the existence proof does not cover equality. For $\sigma = 1$, and for the constructed complementary-input family, $\bar{r} = \rho + \gamma$ instead remains independent of the frontier.

The unbounded limit in (55) compares different economies after taking each economy's long-run limit. Every fixed finite frontier yields finite limiting interest and growth rates. The result establishes neither a finite-time singularity nor an equilibrium in the uncapped economy. The permitted initial-stock neighborhood also depends on the frontier; the comparison does not prove that every frontier admits equilibrium from the same initial stocks.

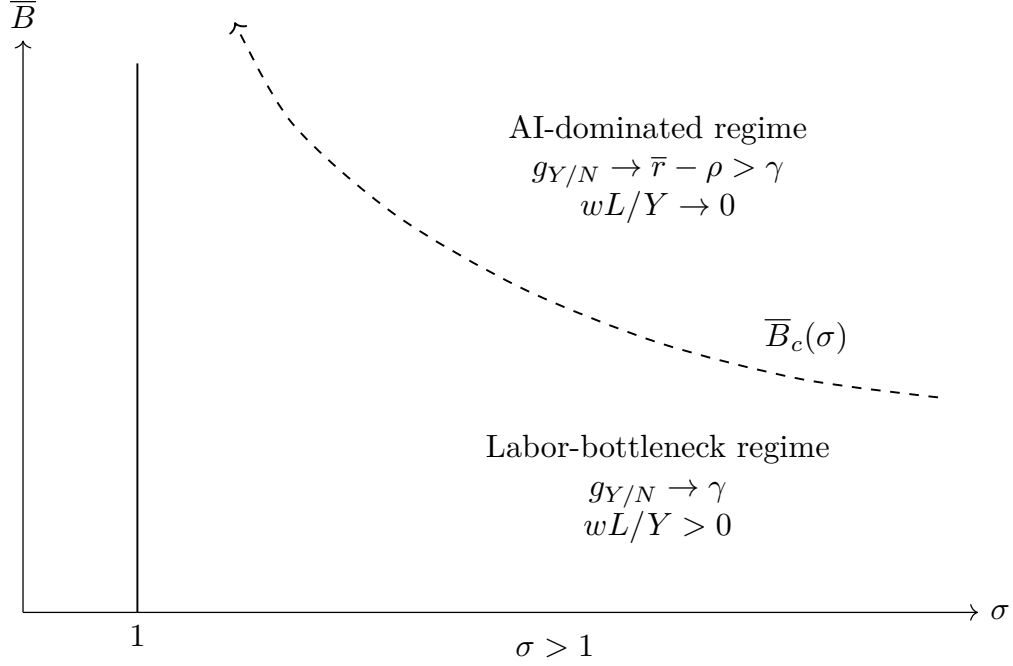


Figure 3: Finite-frontier regimes under gross substitution. The vertical axis is linear and the curve is schematic rather than calibrated. For $\sigma > 1$, $\bar{B}_c(\sigma)$ separates the labor-bottleneck and AI-dominated equilibria constructed in Proposition 5. The dashed boundary itself is not covered by the existence proof.

4.5 Scope of the existence results

Figure 3 summarizes the central comparison for $\sigma > 1$. Here the finite frontier is an analytical device that bounds recursive AI improvement. A frontier below $\bar{B}_c(\sigma)$ supports the labor-bottleneck equilibrium: output per person and the real wage grow at γ , and labor retains a positive income share. Above the threshold, capital and effective AI production services grow faster than effective labor. Output per person then grows faster than γ and labor's income share converges to zero. The figure omits $\sigma < 1$ because I do not interpret $\bar{B}_c(\sigma)$ as a regime boundary under complementarity.

Table 1 collects the sufficient frontier conditions. Every row also requires the common parameter restrictions and the corresponding initial-stock neighborhood. These are not necessary conditions for all possible equilibria, nor a claim of global uniqueness.

Table 1: Constructed finite-frontier equilibria			
Elasticity	Frontier condition	$\lim(g_Y - n)$	$\lim wL/Y$
$0 < \sigma < 1$	$\bar{B} > \bar{B}_c(\sigma)$	γ	Positive
$\sigma = 1$	Any finite $\bar{B} > 0$	γ	$(1 - \alpha)\omega_L$
$\sigma > 1$	$\bar{B} < \bar{B}_c(\sigma)$	γ	Positive
$\sigma > 1$	$\bar{B} > \bar{B}_c(\sigma)$	$\bar{r} - \rho > \gamma$	Zero

The proof constructs convergent trajectories from nearby predetermined stocks and then verifies

both agents' optimality and both transversality conditions. Research remains positive at every finite date. The constants and neighborhoods depend on the parameters; the result does not establish equilibrium from arbitrary K_0, B_0, A_0, N_0 , at the threshold, or after taking $\bar{B} \rightarrow \infty$. Those are separate questions.

The analytical limits do not require simulations. They do not, however, determine transition duration or the paths of wages, interest rates, and income shares. Claims about a prolonged moderate-growth phase require additional analysis of equilibrium transitions.

5 Quantitative equilibrium paths

5.1 Parameters and initial conditions

I compare four economies with the same parameters and predetermined initial stocks, varying only σ . Table 2 retains the macroeconomic benchmarks used in the earlier specification. The AI parameters are illustrative, not estimated. All four economies have autonomous research and a positive AI weight; the no-AI economy in Remark 2 is not one of these four scenarios.

Table 2: Parameters and scenario choices

Parameter	Value	Interpretation	Source or rationale
α	0.33	Capital elasticity	Standard one-third macro benchmark
δ	0.05	Annual capital depreciation	Annual benchmark; PWT comparison (Feenstra et al., 2025)
ρ	0.04	Annual household discount rate	Preference calibration
n	0.003	Annual population growth	Constant-rate approximation to the UN 2024–2100 world projection (United Nations, 2024)
γ	0.01	Annual growth of labor-augmenting technology	Illustrative long-run trend; see OECD (2025)
ω_X	0.20	Weight on effective AI production services	Scenario choice, not an observed industry share
ω_L	0.80	Weight on effective labor	$1 - \omega_X$
η	0.20	Research exponent	$\eta < \alpha$ and $2\eta \leq 1$
χ	0.01	Research productivity	Illustrative research-speed parameter
σ	0.90; 1; 1.10; 1.50	AI–labor substitution	Four scenarios spanning the analytical regimes
$\bar{B}$	$1.10 \bar{B}_c(1.50) \simeq 170.124$	Common AI-efficiency frontier	Ten percent above the threshold for $\sigma = 1.50$

The constant n extrapolates a finite-period demographic approximation; it is not a claim that the UN projects constant population growth forever. The OECD benchmark for γ concerns measured labor-productivity growth, not A directly. I use it only as an approximate guide to the magnitude of long-run labor-augmenting technological progress. The parameter χ determines research speed relative to the other annual rates, so it is not merely a free time-unit normalization. The horizontal axes below measure model years, not calendar forecasts. Likewise, the OECD’s discussion of AI-investment measurement supports treating ω_X as a scenario parameter, but does not identify its value (Fonteneau et al., 2025; OECD, 2026).

The common frontier is below $\bar{B}_c(1.10) \simeq 60.4$ million but above $\bar{B}_c(1.50) \simeq 154.658$. Thus $\sigma = 1.10$ retains a positive long-run labor share, whereas $\sigma = 1.50$ enters the AI-dominated regime. Changing the frontier separately across scenarios would confound this comparison of elasticities.

I set

$$A_0 = N_0 = 1, \quad K_0 = 2.027733654, \quad B_0 = 0.443670932. \quad (56)$$

These stocks come from the unit-elastic balanced-growth reference of the uncapped specification with the same parameters. They are not a balanced-growth state of the capped economy. An exact constant-efficiency reference at $B_0 = \bar{B}$ would eliminate further improvements in AI efficiency and would lie outside the interior problem studied here. In each capped economy, the BVP selects C_0 and q_0 anew; neither jump variable is fixed at its uncapped reference value.

5.2 Numerical method and equilibrium admission

I retain the four-dimensional boundary-value architecture: two initial stock conditions and two terminal conditions selecting the stable equilibrium continuation. The appropriate finite-frontier regime from Section 4, not the uncapped unit-elastic BGP, determines the terminal conditions. Appendix B documents the continuation, equation reconstruction, and numerical checks.

In economic terms, the algorithm takes K_0 and B_0 as given and selects the initial consumption and shadow value that place the economy on the stable equilibrium path. It first constructs that path near the analytically characterized terminal regime and then moves it gradually to the prescribed initial stocks. Solving the boundary-value problem is only the construction step; the separate admission tests determine whether the resulting candidate is reported as an equilibrium.

A converged BVP is not sufficient for admission. I also check feasibility, independently reconstructed dynamic residuals, stability to longer horizons and tighter tolerances, both transversality conditions, and sufficient global developer optimality. The first three scenarios satisfy the global concavity test. For $\sigma = 1.50$, that stronger test fails at low counterfactual AI-efficiency levels, but the Hamiltonian support test in Proposition 9 passes. Its analytical long-run bound is given in (75).

5.3 Growth, wages, and the distribution of output

Figures 4 and 5 present the six main outcomes in two thematic groups. Each figure includes all four admitted trajectories, so the contrast between the labor-bottleneck and AI-dominated regimes remains visible on a common scale. The three labor-bottleneck paths consequently appear close to one another. This compression reflects their common long-run regime; exact initial and limiting values are reported in Table 3.

Panels A and B of Figure 4 report the annual growth rates of output per person and the real wage on the same vertical scale. Their vertical distance measures the growth wedge in Proposition 6. Panel C reports the net interest rate. Figure 5 connects these outcomes to the approach of AI efficiency toward its frontier and to the distribution of final output. AI-industry revenue $p_X X/Y$ is distinct from profit and from value added. Gross capital income remains αY throughout, so the labor and AI-revenue shares sum to $1 - \alpha$.

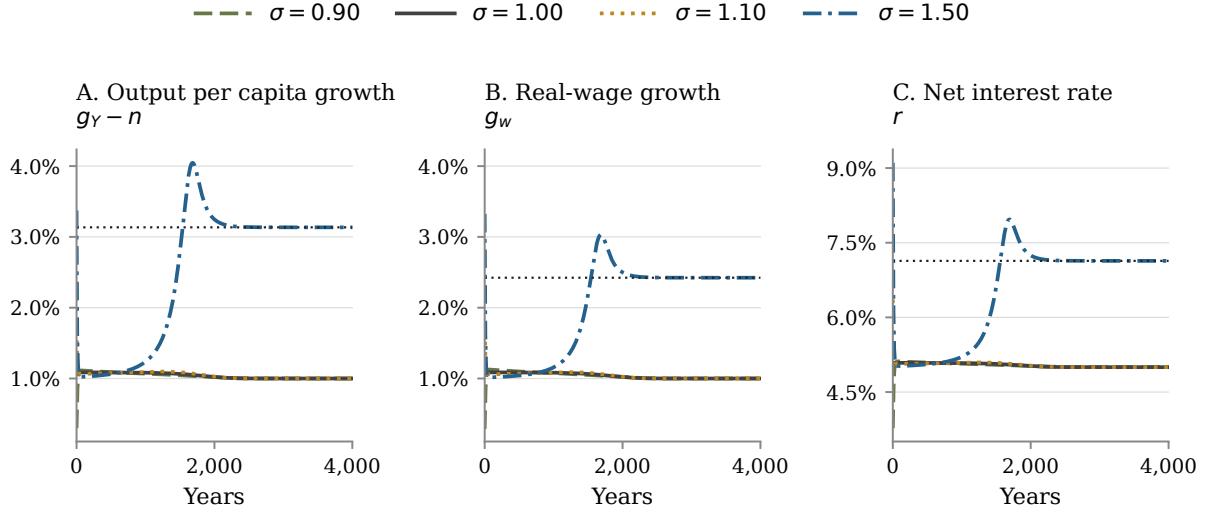


Figure 4: Growth and returns. Each panel compares the same four admitted numerical equilibrium trajectories over 4,000 model years. Output-per-capita growth and real-wage growth use the same linear percentage scale; net interest uses a separate linear percentage scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$. All scenarios have the same initial stocks, frontier, and parameters except σ .

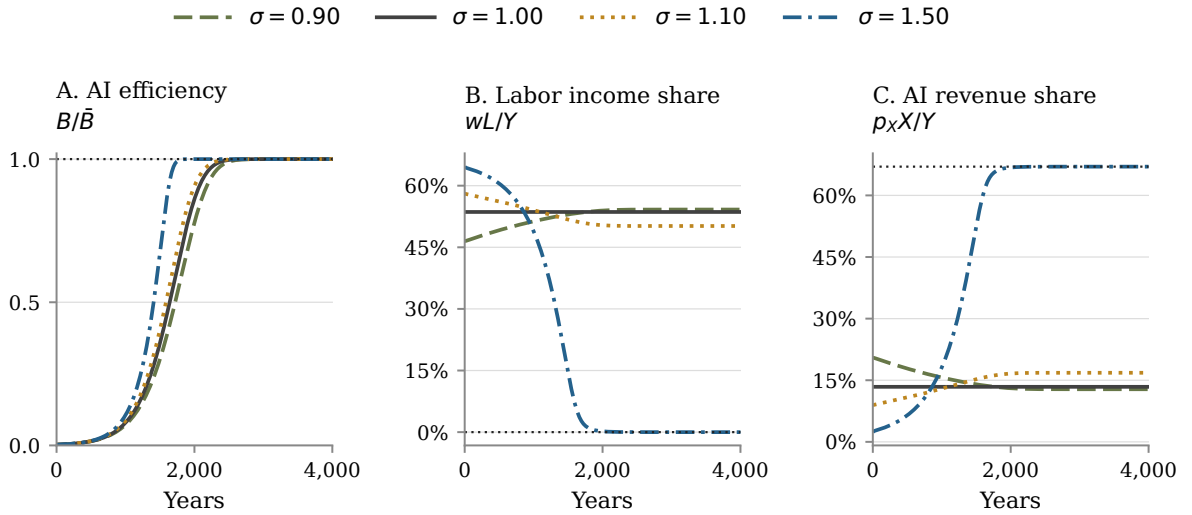


Figure 5: AI transition and the distribution of final output. Each panel compares the same four admitted numerical equilibrium trajectories over 4,000 model years. Panel A reports AI efficiency relative to its frontier; Panels B and C report labor income and AI-industry revenue as shares of final output. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$. All three panels use linear scales.

Table 3: Initial and long-run values of the central outcomes

σ	Date	$g_Y - n$	g_w	r	wL/Y	$p_X X/Y$
0.90	0	0.297	0.288	3.773	46.5	20.5
	∞	1.000	1.000	5.000	54.2	12.8
1.00	0	1.087	1.087	5.087	53.6	13.4
	∞	1.000	1.000	5.000	53.6	13.4
1.10	0	1.831	1.815	6.358	58.1	8.9
	∞	1.000	1.000	5.000	50.2	16.8
1.50	0	3.373	3.326	9.107	64.6	2.4
	∞	3.135	2.423	7.135	0.0	67.0

Note: All entries are percentages. Date-zero entries come from the numerical equilibrium paths. Infinite-date entries are the analytical limits characterized in Section 4.

The first three scenarios approach per-person growth of 1% and a net interest rate of 5%. For $\sigma = 1.50$, their analytical limits are 3.135% and 7.135%, respectively. Real wages then grow asymptotically at 2.423%. Thus output per person grows 0.712 percentage points faster than the real wage. By Proposition 6, this implies an asymptotic decline of 0.712% per year in the log labor income share. The central distributional result is therefore not that real wages decline: workers become richer in absolute terms while their claim on total output vanishes. AI revenue meanwhile approaches 67% of output. These limiting rates describe the equilibrium continuations; they are not imposed as constant rates during the transition.

The transition matters. At $\sigma = 1.50$, the initial adjustment raises growth and interest temporarily; both then fall close to the labor-supported rates. At model year 500, per-person growth is still only 1.062% and labor receives 60.5% of output. By year 1,500, growth reaches 2.797% and the labor share falls to 14.5%. Growth and interest briefly exceed their eventual limits before converging. Thus a long interval of moderate growth can precede a sharp change in both prosperity and income shares. Its duration depends on the illustrative research calibration, not on an estimated date for an AI transition.

Substitution alone does not produce this outcome. With the same frontier and $\sigma = 1.10$, the labor share instead approaches 50.2%, compared with 53.6% at $\sigma = 1$ and 54.2% at $\sigma = 0.90$. The frontier's position relative to the elasticity-specific threshold is essential to interpreting the comparison.

5.4 Technology, accumulation, and AI-revenue composition

Figure 6 reports consumption and capital per unit of effective labor and the use of AI-industry revenue. The last three panels deliberately divide by $p_X X$, not by Y :

$$\frac{U}{p_X X} + \frac{M}{p_X X} + \frac{\Pi}{p_X X} = 1. \quad (57)$$

Figure 5 measures the industry's size relative to final output; Figure 6 separates its inference costs, research expenditure, and net distribution to its owner.

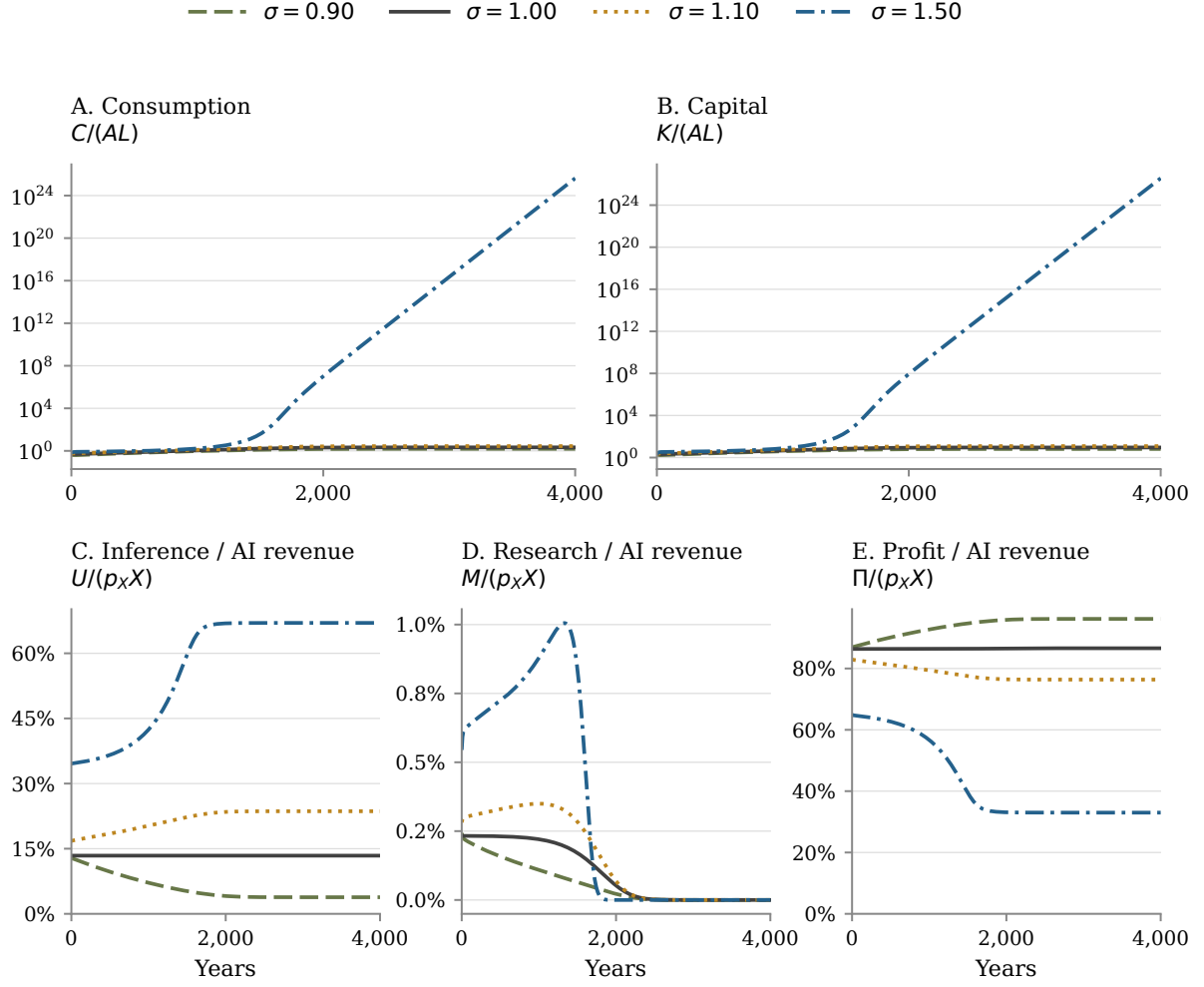


Figure 6: Accumulation and the composition of AI revenue. The scenarios and time window match Figure 4. Consumption and capital per unit of effective labor use logarithmic scales. The three revenue shares use linear scales and sum to one; negative net distributions, if any, represent household equity injections rather than negative gross revenue.

In the AI-dominated limit, inference absorbs 67% of AI revenue, research's revenue share vanishes, and the developer retains 33%. Its net distribution therefore approaches 22.11% of final output. By contrast, at $\sigma = 1$ AI revenue is always 13.4% of output and inference absorbs 13.4% of that revenue. A vanishing research share does not mean that research stops: the constructed finite-frontier equilibria have a positive limiting level of research compute, while industry revenue continues to grow.

6 Conclusion

The endogenous growth model showed that the long-run economic role of human labor depends critically on whether AI can replace labor’s contribution to production. It characterizes how this substitution margin shapes economic growth, wages, interest rates, and the functional distribution of income when AI research is autonomous.

When AI and labor are complements, human labor remains a bottleneck. Improvements in AI efficiency raise productive capacity but do not permanently increase the growth of output per person or real wages beyond the exogenous rate of labor-augmenting technological progress. Labor therefore retains a positive share of income. The same broad conclusion holds under unit elasticity: AI can support sustained growth, but human labor remains economically essential and continues to receive a positive share of output.

The outcome changes when AI and labor are sufficiently substitutable. Gross substitution is necessary but not sufficient: labor remains a bottleneck while AI services are not productive enough to replace its contribution. Once AI becomes sufficiently efficient, however, capital and effective AI production services can expand faster than effective labor. Output per person, real wages, and the interest rate then grow faster than in the labor-bottleneck regime.

The distributional result is not that workers necessarily become poorer. In the AI-dominated regime, real wages continue to rise, but output per person grows even faster. Labor’s income share therefore converges to zero. Because the gross capital share is fixed by the production technology, this transformation reallocates the remainder of output from labor toward the AI industry. AI-industry revenue should not be confused with monopoly profit: it finances inference and research before the residual is distributed to the developer’s owner.

The transition can be as important as the long-run outcome. The numerical equilibrium paths show that the economy may remain for a long time on a moderate-growth trajectory before output accelerates and labor’s income share falls sharply. A prolonged period in which little appears to change therefore need not imply that labor will remain economically essential. The timing in the simulations is illustrative, however, because the speed of AI research is not estimated.

The analysis points to three lines of future work. A task-based extension with heterogeneous workers, new human tasks, and a continuing role for human researchers could combine the displacement and reinstatement mechanisms of [Acemoglu and Restrepo \(2019\)](#) with the research-automation network of [Davidson et al. \(2026\)](#). An explicit upstream compute sector, alternative market structures, and heterogeneous asset ownership could build on [Korinek and McKelvey \(2026\)](#) and [Korinek and Vipra \(2024\)](#) and translate changes in functional income shares into household inequality and economic power. Finally, a more complete existence analysis could replace bounded AI efficiency with an endogenous or unbounded process and determine whether the resulting economy admits an infinite-horizon equilibrium. This question is related to, but not resolved by, the explosive-growth mechanisms in [Nordhaus \(2021\)](#) and [Davidson et al. \(2026\)](#).

The central question is whether human labor remains sufficiently essential for wages to keep pace with the economy as a whole.

A Proofs

A.1 Finite-frontier invariance

Let $D = \bar{B} - B$. The finite-frontier state equation gives

$$\frac{\dot{D}}{D} = -\frac{\chi}{B}(BM)^\eta.$$

Integration gives

$$\bar{B} - B_t = (\bar{B} - B_0) \exp \left\{ -\frac{\chi}{B} \int_0^t (B_s M_s)^\eta ds \right\} > 0. \quad (58)$$

Local integrability makes the exponent finite at every finite date, so $D_t > 0$.

A.2 Existence and uniqueness of the static service choice

Fix $K, A, L, B > 0$, let $R(X) = p_X(X)X$, and write

$$u = \frac{1}{\sigma}, \quad s = s_X, \quad e = u(1 - s) + \alpha s.$$

The demand elasticity in (13) and the CES-share derivative imply

$$R'(X) = p_X(1 - e), \quad \frac{ds}{d \ln X} = (1 - u)s(1 - s).$$

Let $m(X) = R'(X)$. Wherever marginal revenue is positive, direct differentiation gives

$$-\frac{Xm'(X)}{p_X} = e(1 - e) + (\alpha - u)(1 - u)s(1 - s) \equiv D(s). \quad (59)$$

If $(\alpha - u)(1 - u) \geq 0$, then $D(s) > 0$ immediately. If this product is negative, the parameter restrictions imply $0 < \alpha < u < 1$, and

$$D(s) = u(1 - u)(1 - s)^2 + [u^2 + 2\alpha - 3\alpha u]s(1 - s) + \alpha(1 - \alpha)s^2.$$

The middle coefficient is positive. For $u \leq 2/3$, this follows from $u^2 + \alpha(2 - 3u) > 0$. For $u > 2/3$, use $\alpha < u$ and $2 - 3u < 0$ to obtain

$$u^2 + \alpha(2 - 3u) > u^2 + u(2 - 3u) = 2u(1 - u) > 0.$$

Thus marginal revenue decreases strictly whenever it is positive.

For $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$. As $X \downarrow 0$, revenue is proportional to X^{1-u} when $\sigma > 1$, to X^β when $\sigma = 1$, and to $X^{1-\alpha}$ when $\sigma < 1$. Every exponent lies between zero and one, so $m(X) \rightarrow \infty$. As $X \rightarrow \infty$, marginal revenue converges to zero from above if $\sigma \geq 1$; if $\sigma < 1$, it eventually becomes negative. Hence $m(X) = 1/B$ has exactly one solution. Operating profit increases before this

solution and decreases after it. The linear cost X/B dominates revenue as $X \rightarrow \infty$, so this solution is the unique global maximum. Equation (59) also proves the strict second-order condition:

$$e_X(1 - e_X) + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_X(1 - s_X) > 0. \quad (60)$$

□

A.3 Interior research

Appendix A.2 establishes $X^* > 0$, and the envelope condition (23) gives $\pi_B = X^*/B^2 > 0$. A marginal increase in AI efficiency therefore raises current optimized profit and weakly raises future AI efficiency under a fixed research policy. At a differentiable optimum its current-value shadow price is strictly positive: $q > 0$. The part of the Hamiltonian that depends on M is

$$-M + q\chi B^\eta M^\eta \psi(B).$$

Its right derivative diverges as $M \downarrow 0$ because $q, B, \psi(B) > 0$ and $0 < \eta < 1$. Therefore $M = 0$ cannot maximize the Hamiltonian at a finite date. □

A.4 Research expenditure and finite-horizon value

The first-order and transversality conditions do not by themselves establish that the developer's objective has a finite value or that a candidate policy is globally optimal. The result below addresses the value on a fixed finite horizon, taking the paths of K, A, L , and r as given. To explain the role of Assumption 1, I also consider values of η outside its maintained range, retaining $0 < \eta < 1$ from the AI-efficiency law. Global optimality is addressed separately in Proposition 8.

Proposition 7 (Large-scale research expenditure). *Fix a finite horizon, an initial AI-efficiency level, positive continuous paths for K, A, L , and an integrable path for r . In the uncapped economy with $\bar{B} = \infty$, if $0 < \eta < \alpha < 1$, the developer's truncated value is finite and its objective tends to minus infinity as total research expenditure diverges. If $\sigma > 1$ and $\eta > \alpha$, fixed-duration expansions of research compute make the truncated value unbounded above. At $\eta = \alpha$, the leading benefit and cost terms have the same order, so their coefficients determine the result. For a fixed finite frontier, the truncated problem instead has a finite value and attains its maximum for every $0 < \eta < 1$.*

The uncapped result compares the cost of research with its effect on future operating profit. Expenditure first raises AI efficiency; higher AI efficiency then raises the profit from supplying effective AI production services. The proof bounds these two effects in turn and compares their combined growth with the linear cost of research.

Proof. I first consider the uncapped economy. Define

$$\theta = \frac{1-\alpha}{\alpha}, \quad S = \int_0^T M_t dt.$$

Here S is total research expenditure over the fixed horizon $[0, T]$. For an admissible research policy, let its discounted net payoff be

$$J_T(M) = \int_0^T \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt.$$

For every admissible policy, B is nondecreasing. The AI-efficiency law implies

$$\frac{d}{dt} B_t^{1-\eta} = (1-\eta)\chi M_t^\eta.$$

Integration and Hölder's inequality give

$$\begin{aligned} B_t^{1-\eta} &= B_0^{1-\eta} + (1-\eta)\chi \int_0^t M_s^\eta ds, \\ \int_0^T M_t^\eta dt &\leq T^{1-\eta} S^\eta. \end{aligned}$$

Because $0 < \eta < 1$, spreading a given expenditure evenly over $[0, T]$ maximizes the integral of M^η . Hence there is a constant $C_B > 0$, independent of the research policy, such that

$$\max_{0 \leq t \leq T} B_t \leq C_B \left(1 + S^{\eta/(1-\eta)} \right), \quad (61)$$

The same bound covers arbitrarily concentrated controls because, on a support of length $d \leq T$, $\int M_t^\eta dt \leq d^{1-\eta} S^\eta$.

Let $\bar{K} = \max_{t \leq T} K_t$ and $\bar{\mathcal{L}} = \max_{t \leq T} A_t L_t$. The weighted power-mean inequality gives $Z \leq AL + X$. The final-good firm's demand condition therefore implies

$$p_X X \leq (1-\alpha) \bar{K}^\alpha \left(\bar{\mathcal{L}}^{1-\alpha} + X^{1-\alpha} \right).$$

Write the right-hand side as $a + bX^{1-\alpha}$, with $a, b > 0$ determined by the given paths and production parameters. The first-order condition for maximizing this revenue bound net of X/B is $b(1-\alpha)X^{-\alpha} = 1/B$, and its maximized value is

$$\max_{X \geq 0} \left\{ a + bX^{1-\alpha} - \frac{X}{B} \right\} = a + \alpha b [b(1-\alpha)B]^\theta.$$

There is therefore a constant $C_\pi > 0$, independent of the research policy, such that

$$\pi(B) \leq C_\pi (1 + B^\theta). \quad (62)$$

Combining the AI-efficiency and profit bounds shows that operating profit grows no faster than a constant plus a term proportional to

$$S^{\eta(1-\alpha)/[\alpha(1-\eta)]}, \quad (63)$$

where the exponent is the product of $\eta/(1-\eta)$, from expenditure to AI efficiency, and $(1-\alpha)/\alpha$, from AI efficiency to profit. This is an upper bound on the order of growth, not an exact profit function.

On a finite horizon, integrability of r bounds the discount factor above and away from zero. Let $\underline{D} > 0$ be its minimum on $[0, T]$. Combining (61) and (62) yields, for constants $C_0, C_1 > 0$ independent of the research policy,

$$J_T(M) \leq C_0 + C_1 S^p - \underline{D}S, \quad p = \frac{\eta(1-\alpha)}{\alpha(1-\eta)}. \quad (64)$$

The exponent satisfies $p < 1$ exactly when $\eta < \alpha$. In this case, the benefit grows at most sublinearly in expenditure while its cost grows linearly. The bound tends to minus infinity as $S \rightarrow \infty$, so the value is finite and the objective is coercive.

An upper bound with $p > 1$ does not by itself prove unboundedness. For the converse, I construct research policies with unbounded payoffs. Suppose $\sigma > 1$. As $B \rightarrow \infty$, optimized operating profit satisfies

$$\pi(B) \sim \kappa_X K B^\theta$$

for a constant $\kappa_X > 0$, uniformly when K, A , and L remain in compact subsets of the positive orthant. Fix $\Delta \in (0, T)$, choose $M = \mathcal{B}$ on $[0, \Delta]$, and choose $M = 0$ afterward. Then

$$B_{\mathcal{B}}(t)^{1-\eta} = B_0^{1-\eta} + (1-\eta)\chi \mathcal{B}^\eta \min\{t, \Delta\}.$$

Choosing X optimally at every date gives

$$J_T(\mathcal{B}) = c_\pi \mathcal{B}^{\eta(1-\alpha)/[\alpha(1-\eta)]} - c_M \mathcal{B} + o\left(\mathcal{B}^{\eta(1-\alpha)/[\alpha(1-\eta)]}\right), \quad (65)$$

with $c_\pi, c_M > 0$. The exponent exceeds one when $\eta > \alpha$, so this family makes the truncated value unbounded above. At equality, the exponent is one and the sign of the leading coefficient determines this family's result.

For a finite frontier, the invariance result above confines B to the compact interval $[B_0, \overline{B}]$. Optimized operating profit is continuous in B and uniformly bounded on the fixed horizon. Hence $J_T(M) \leq C_0 - \underline{D}S$, so every maximizing sequence has bounded total expenditure. Set $v = M^\eta$ and $p = 1/\eta > 1$. The control cost is v^p , while the state equation is affine in v :

$$\dot{B} = \chi B^\eta \left(1 - \frac{B}{\overline{B}}\right) v.$$

A maximizing sequence is bounded in the reflexive space $L^p[0, T]$. A weakly convergent subsequence

and the associated uniformly convergent state paths preserve feasibility. The operating-profit integral is continuous under uniform state convergence, while $\int v^p$ is weakly lower semicontinuous. The objective is therefore weakly upper semicontinuous and attains its maximum. $\square$

Assumption 1 prevents arbitrarily large research expenditure from being optimal on a fixed horizon in the uncapped economy. It is a sufficient restriction, not a necessary one for every value of σ , and finite-horizon existence with a finite frontier does not require $\eta < \alpha$. The restriction supplies a common parameter range that also covers the uncapped research technology. Neither the bound nor finite-horizon developer existence establishes an infinite-horizon equilibrium: the paths of K, A, L , and r were taken as given throughout this argument. The quantitative analysis uses $\eta = 0.20 < \alpha = 0.33$, for which the expenditure exponent is approximately 0.508; the value of η is illustrative rather than estimated.

A.5 Global optimality of the developer's policy

To state a sufficient condition for global optimality, define the service-efficiency elasticity

$$\varepsilon_B \equiv \left. \frac{\partial \ln X^*(B)}{\partial \ln B} \right|_{K,A,L} = \frac{1 - e_X}{e_X(1 - e_X) + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_X(1 - s_X)}. \quad (66)$$

Because $\pi_B = X^*/B^2$,

$$\pi_{BB} = \frac{X^*}{B^3}(\varepsilon_B - 2). \quad (67)$$

For a finite frontier, define research depth

$$R_B = -\bar{B} \log \left(1 - \frac{B}{\bar{B}} \right). \quad (68)$$

Then $B(R_B) = \bar{B}[1 - e^{-R_B/\bar{B}}]$ and

$$\dot{R}_B = \chi[B(R_B)M]^\eta. \quad (69)$$

A direct differentiation gives

$$\frac{\partial^2 \pi(B(R_B))}{\partial R_B^2} = \frac{X\psi(B)}{B^3} \left[\psi(B)(\varepsilon_B - 2) - \frac{B}{\bar{B}} \right]. \quad (70)$$

Proposition 8 (Developer sufficiency with a finite frontier). *Consider a feasible finite-frontier developer policy with a finite objective. Suppose it satisfies the pointwise monopoly choice, research first-order condition, AI-efficiency law, costate equation, and developer transversality condition derived above. If $2\eta \leq 1$ and*

$$\psi(B)(\varepsilon_B - 2) \leq \frac{B}{\bar{B}} \quad (71)$$

for every date and every AI-efficiency level in the reachable interval $[B_0, \bar{B}]$, then the policy is a

global solution to the developer's problem.

Condition (71) is sufficient, not necessary. It holds strictly near the frontier because its left-hand side converges to zero while its right-hand side converges to one. It can fail at lower AI-efficiency levels, which is why the numerical equilibrium audit checks it along the entire reachable path.

Proof. The transformation (68) satisfies

$$B'(R_B) = \psi(B) > 0, \quad B''(R_B) = -\frac{\psi(B)}{B} < 0.$$

Using $\pi_B = X/B^2$ and (67) gives (70). Condition (71) therefore makes optimized operating profit concave in R_B throughout the reachable state domain.

When $2\eta \leq 1$, the monomial $(BM)^\eta$ is jointly concave and increasing in (B, M) . Because $B(R_B)$ is increasing and concave, $\chi[B(R_B)M]^\eta$ is jointly concave in (R_B, M) . Research cost is linear. The present-value Hamiltonian written in research depth is therefore jointly concave, and the transformed first-order and costate conditions maximize it globally.

The shadow value of research depth is $q_R = qB'(R_B) = q\psi(B)$. Moreover,

$$\psi(B)R_B = -(\bar{B} - B) \log \left(1 - \frac{B}{\bar{B}} \right) \leq B.$$

The developer's transversality condition therefore implies the terminal condition required by the concavity argument in R_B . For any admissible alternative with the same initial stock, the standard finite-horizon concavity inequality gives

$$J_T^* - \tilde{J}_T \geq \mu_T(\tilde{R}_{B,T} - R_{B,T}^*) \geq -\mu_T R_{B,T}^*,$$

where μ_T is the present-value shadow price of research depth. The final term converges to zero. Taking the lower limit as $T \rightarrow \infty$ establishes global optimality. $\square$

A sufficient condition without global concavity. The preceding curvature condition can fail at counterfactual AI-efficiency levels even when the candidate is globally optimal. An alternative is to check that the maximized Hamiltonian lies below its tangent at the candidate state. This changes the optimality test, not the developer's problem.

Fix a date, its capital and effective labor, and the candidate shadow price $q_{R,t} = q_t\psi(B_t) > 0$. Maximizing the research-depth Hamiltonian over research expenditure gives

$$\hat{\mathcal{H}}_t(\tilde{R}_B) = \pi(B(\tilde{R}_B)) + \frac{1-\eta}{\eta} M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)}. \quad (72)$$

Here M_t is the candidate's research expenditure. The formula follows by maximizing $-M + q_{R,t}\chi[B(\tilde{R}_B)M]^\eta$ over $M \geq 0$ and using the candidate's research first-order condition. Its derivative

at the candidate state is

$$\widehat{\mathcal{H}}'_t(R_{B,t}) = \frac{\psi(B_t)}{B_t}(U_t + M_t). \quad (73)$$

Proposition 9 (Global Hamiltonian support). *Consider a feasible finite-frontier developer policy with a finite objective, positive shadow price, and $0 < \eta < 1$. Suppose it satisfies the monopoly choice, research first-order condition, AI-efficiency law, costate equation, and developer transversality condition. It is globally optimal if, at every date and for every $\widetilde{R}_B \geq R_{B,0}$,*

$$\widehat{\mathcal{H}}_t(\widetilde{R}_B) \leq \widehat{\mathcal{H}}_t(R_{B,t}) + \widehat{\mathcal{H}}'_t(R_{B,t})(\widetilde{R}_B - R_{B,t}). \quad (74)$$

Proof. Write $D_t = \exp(-\int_0^t r_s ds)$ and $\mu_t = D_t q_{R,t}$. The transformed costate equation is $\dot{\mu}_t = -D_t \widehat{\mathcal{H}}'_t(R_{B,t})$. For any admissible alternative research policy and its state $\widetilde{R}_{B,t}$, maximizing over its control first bounds its Hamiltonian by (72). Apply (74), subtract the candidate payoff, and integrate by parts. Equal initial states imply

$$\widetilde{J}_T - J_T \leq -\mu_T(\widetilde{R}_{B,T} - R_{B,T}) \leq \mu_T R_{B,T}.$$

Because $\psi(B_t)R_{B,t} \leq B_t$, the last expression converges to zero by the developer's transversality condition. Taking the upper limit establishes global optimality. The argument requires a global upper tangent at the candidate state, not concavity at every counterfactual state. $\square$

The AI-dominated continuation supplies an analytical long-run check on the support inequality. For $1 < \sigma \leq 1/\alpha$, the service elasticity satisfies $\varepsilon_B \leq 1/\alpha$. Indeed, subtracting $\alpha(1 - e_X)$ from the denominator in (66) gives

$$(\sigma^{-1} - \alpha)(1 - s_X) [(1 - \sigma^{-1})(1 - s_X) + (\sigma^{-1} - \alpha)s_X] \geq 0.$$

Choose $B_* \in (B_0, \overline{B})$ with $B_*/\overline{B} > \max\{0, (1 - 2\alpha)/(1 - \alpha)\}$. Operating profit is then concave in research depth for $B \geq B_*$. When $2\eta \leq 1$, the research term in (72) is concave there as well. For a candidate with $B_t \geq B_*$, it consequently suffices to check

$$\widehat{\mathcal{H}}_t(R_{B,t}) - \widehat{\mathcal{H}}'_t(R_{B,t})(R_{B,t} - R_{B,0}) \geq \widehat{\mathcal{H}}_t(R_B(B_*)). \quad (75)$$

Above B_* the tangent bound follows from concavity. Below B_* the Hamiltonian is increasing in B , so its value is at most the right-hand side, whereas the tangent is minimized at $R_{B,0}$.

Along an AI-dominated continuation, the difference between the two sides of (75), divided by Y_t , converges to

$$\alpha(1 - \alpha) \left[1 - \left(\frac{B_*}{\overline{B}} \right)^{(1-\alpha)/\alpha} \right] > 0.$$

To obtain this limit, use $M/Y \rightarrow 0$, $\psi(B)R_B \rightarrow 0$, and $\pi(B_*)/Y \rightarrow \alpha(1 - \alpha)(B_*/\overline{B})^{(1-\alpha)/\alpha}$. Thus the global support condition holds sufficiently late on this continuation even when the stronger

curvature condition fails at low counterfactual AI-efficiency levels. The finite transition still requires its own support check.

A.6 Finite-frontier equilibrium existence

This subsection proves Propositions 3–5. I first construct the terminal values, then a family of convergent solutions of the exact equilibrium equations, and finally verify optimality and transversality. No equilibrium trajectory or convergence assumption is used as a premise.

A.6.1 Static allocations and the frontier threshold

Write $\theta = (1 - \alpha)/\alpha$. For $\sigma \neq 1$, let $\varphi = (\sigma - 1)/\sigma$ and define

$$m(s) = 1 - \frac{1-s}{\sigma} - \alpha s, \quad \mathcal{H}_\sigma(s) = \left[(1 - \alpha)s m(s) \left(\frac{\omega_X}{s} \right)^{1/\varphi} \right]^\theta.$$

Here $m(s) = 1 - e_X$ is the marginal-revenue factor, not research compute. The static monopoly optimum has $m(s_X) > 0$ by Appendix A.2. The CES share, monopoly condition, and production function imply

$$\frac{X}{AN} = \left[\frac{\omega_L s_X}{\omega_X (1 - s_X)} \right]^{1/\varphi}, \quad \frac{U}{Y} = (1 - \alpha)s_X m(s_X), \quad (76)$$

$$\frac{Y}{K} = B^\theta \mathcal{H}_\sigma(s_X). \quad (77)$$

For example, $Z = X(\omega_X/s_X)^{1/\varphi}$ and $X = B(1 - \alpha)s_X m(s_X)Y$ give the last identity directly.

If $0 < \sigma < 1$, positive marginal revenue requires

$$s > s_{\min} \equiv \frac{1 - \sigma}{1 - \alpha\sigma}.$$

On $(s_{\min}, 1)$, both $m(s)$ and $s^{-1/(\sigma-1)}$ increase strictly. Since

$$\mathcal{H}_\sigma(s)^{1/\theta} = (1 - \alpha)\omega_X^{1/\varphi} m(s)s^{-1/(\sigma-1)},$$

the share map increases from zero to $\mathcal{H}_\sigma(1) = [(1 - \alpha)^2 \omega_X^{1/\varphi}]^\theta$. Thus the equation

$$\mathcal{R} = \overline{B}^\theta \mathcal{H}_\sigma(\overline{s}_X) \quad (78)$$

has exactly one interior solution if and only if $\overline{B} > \overline{B}_c(\sigma)$.

If $\sigma > 1$, the share map decreases strictly on $(0, 1)$. To verify the sign, put $a = \sigma^{-1} - \alpha$, so $m(s) = \varphi + as$. Its logarithmic derivative has the sign of $a(\sigma - 2)s - \varphi$. If $a(\sigma - 2) \leq 0$ this is negative. If $\sigma > 2$ and $a > 0$, $a(\sigma - 2) < (\sigma - 2)/\sigma < \varphi$; if $1 < \sigma < 2$ and $a < 0$, $a(\sigma - 2) < (1 - \sigma^{-1})(2 - \sigma) < \varphi$. The map therefore decreases from infinity to $\mathcal{H}_\sigma(1)$. Equation (78) has exactly one interior solution if and only if $\overline{B} < \overline{B}_c(\sigma)$. These calculations also derive (38)–(39).

For either interior-share case, let x, y, u, k, c denote X, Y, U, K, C divided by AN , respectively. Construct the normalized terminal values sequentially:

$$\begin{aligned}\bar{x} &= \left[\frac{\omega_L \bar{s}_X}{\omega_X (1 - \bar{s}_X)} \right]^{1/\varphi}, & \bar{u} &= \frac{\bar{x}}{\bar{B}}, \\ \bar{y} &= \frac{\bar{u}}{(1 - \alpha) \bar{s}_X m(\bar{s}_X)}, & \bar{k} &= \frac{\bar{y}}{\bar{\mathcal{R}}}, \\ \bar{c} &= \bar{y} - \bar{u} - (\delta + n + \gamma) \bar{k}, & \bar{r} &= \rho + \gamma.\end{aligned}\tag{79}$$

At $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$ as in the main text. The exact static solution is

$$y = \left[(\beta^2 B)^\beta k^\alpha \right]^{1/(1-\beta)}, \quad u = \beta^2 y, \quad x = Bu.$$

The equation $y/k = \mathcal{R}$ gives (42). It has exactly one positive solution for every finite $\bar{B} > 0$; set $\bar{y} = \mathcal{R} \bar{k}$, $\bar{u} = \beta^2 \bar{y}$, and $\bar{x} = \bar{B} \bar{u}$, with $\bar{c}, \bar{r}$ as in (79).

Consumption positivity does not require an additional restriction. Positive marginal revenue implies $0 < m(\bar{s}_X) < 1$. Since $0 < \bar{s}_X < 1$, in all three cases

$$\frac{\bar{u}}{\bar{y}} = (1 - \alpha) \bar{s}_X m(\bar{s}_X) < 1 - \alpha, \quad \frac{(\delta + n + \gamma) \bar{k}}{\bar{y}} = \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} < \alpha.$$

It follows that $\bar{c}/\bar{y} > 0$.

A.6.2 Construction near a positive labor-share regime

Use the coordinates

$$k = \frac{K}{AN}, \quad c = \frac{C}{AN}, \quad d = (\bar{B} - B)AN, \quad \tilde{q} = \frac{q}{AN}, \quad \tau = (AN)^{-1}.\tag{80}$$

They leave the model unchanged. Given k, B , solve the unique static monopoly problem for y, x, u , and set $r = \alpha y/k - \delta$. Define $a = \dot{B}/(\bar{B} - B)$, the proportional rate of closure of the AI-efficiency gap. The research condition gives

$$B = \bar{B} - d\tau, \quad M = \left[\tilde{q} d \chi \eta \frac{B^\eta}{\bar{B}} \right]^{1/(1-\eta)}, \quad a = \chi \frac{B^\eta M^\eta}{\bar{B}}.$$

The exact dynamic equations are

$$\frac{\dot{k}}{k} = \frac{y - c - u - M\tau}{k} - \delta - n - \gamma, \quad (81a)$$

$$\frac{\dot{c}}{c} = r - \rho - \gamma, \quad (81b)$$

$$\frac{\dot{d}}{d} = n + \gamma - a, \quad (81c)$$

$$\frac{\dot{\tilde{q}}}{\tilde{q}} = r - \frac{x}{\tilde{q}B^2} - \eta \frac{ad\tau}{B} + a - n - \gamma, \quad (81d)$$

$$\dot{\tau} = -(n + \gamma)\tau. \quad (81e)$$

In particular, population growth cancels from the normalized consumption equation; it is not subtracted a second time.

Complete the terminal construction by setting

$$\bar{M} = \left[\frac{(n + \gamma)\bar{B}^{1-\eta}}{\chi} \right]^{1/\eta}, \quad \bar{\tilde{q}} = \frac{\bar{x}}{\bar{r}\bar{B}^2}, \quad \bar{d} = \frac{\bar{M}}{\eta(n + \gamma)\bar{\tilde{q}}}. \quad (82)$$

All are positive. Direct substitution shows that $(\bar{k}, \bar{c}, \bar{d}, \bar{\tilde{q}}, 0)$ is a fixed point of (81). This boundary point is a device for constructing paths with $\tau > 0$ and $B < \bar{B}$, not an admissible date-zero state itself.

I next verify saddle-path stability. Write $\mu = \eta/(1 - \eta)$, $r_k = \partial r / \partial \log k$, and $a_k = \partial[(y - u)/k] / \partial \log k + \bar{c}/\bar{k}$, evaluated at the fixed point. In $(\log k, \log c, \log d, \log \tilde{q}, \tau)$ coordinates the Jacobian has the two diagonal blocks

$$J_R = \begin{pmatrix} a_k & -\bar{c}/\bar{k} \\ r_k & 0 \end{pmatrix}, \quad J_B = \begin{pmatrix} -(n + \gamma)\mu & -(n + \gamma)\mu \\ (n + \gamma)\mu & \bar{r} + (n + \gamma)\mu \end{pmatrix}, \quad (83)$$

and the additional eigenvalue $-(n + \gamma)$. There can be forcing from τ to both blocks and from (k, c) to $(d, \tilde{q})$, but not in the reverse directions at the fixed point.

To check $r_k < 0$, let $s = s_X$, $u_\sigma = 1/\sigma$, $e = e_X$, and let $D(s)$ be the positive curvature expression in (59). Implicit differentiation of the monopoly condition gives

$$\frac{\partial \log X}{\partial \log K} = \frac{\alpha(1 - e)}{D(s)}, \quad r_k = (r + \delta)(1 - \alpha) \left[\frac{\alpha s(1 - e)}{D(s)} - 1 \right].$$

Moreover,

$$D(s) - \alpha s(1 - e) = (1 - s)\{u_\sigma(1 - e) + (\alpha - u_\sigma)(1 - u_\sigma)s\} > 0.$$

The sign is immediate when $u_\sigma \geq 1$ or $u_\sigma \leq \alpha$. For $\alpha < u_\sigma < 1$, the braces are linear in s , with positive endpoint values $u_\sigma(1 - u_\sigma)$ and $u_\sigma^2 + \alpha(1 - 2u_\sigma)$. Thus $r_k < 0$ in every interior-share case.

Both blocks have negative determinants:

$$\det J_R = (\bar{c}/\bar{k})r_k < 0, \quad \det J_B = -(n + \gamma)\mu\bar{r} < 0.$$

There are therefore three stable and two unstable eigenvalues. The stable direction in J_R has a nonzero k component, and the stable direction in J_B has a nonzero d component. Given the stable τ direction, one can successively eliminate the growing solution in the Ramsey block by choosing c , and then eliminate the growing solution in the research block by choosing $\tilde{q}$. Hence the stable subspace projects invertibly onto (k, d, τ) . This argument does not require distinct stable eigenvalues.

The static solution and the vector field are smooth near the terminal point. The stable-manifold theorem and the inverse-function theorem therefore give a local graph $(c, \tilde{q})$ over (k, d, τ) ; see [Dyatlov \(2018\)](#) for the stable-manifold construction. Every physical initial state in a sufficiently small neighborhood (40) has one nearby convergent solution on this graph. The solution remains in the neighborhood for all future time. Positivity of $\tau, d, c, \tilde{q}, k$ and the original AI-efficiency law ensure $0 < B_t < \bar{B}$ and $M_t > 0$ at every finite date. Reconstruction gives the original initial stocks, the equilibrium equations, and the limiting rates in (41) and (43). Optimality and transversality are verified below.

A.6.3 Construction near the AI-dominated regime

Let $\sigma > 1$ and $\bar{B} > \bar{B}_c(\sigma)$. Here $\varphi = (\sigma - 1)/\sigma \in (0, 1)$. Use the alternative coordinates

$$h = (AN/K)^\varphi, \quad d = (\bar{B} - B)K, \quad c = C/K, \quad v = q/K, \quad \tau = (AN)^{-1}.$$

Within this subsubsection, y, x, u denote $Y/K, X/K, U/K$, respectively. Then

$$B = \bar{B} - d\tau h^{1/\varphi}, \quad M = [vd\chi\eta B^\eta/\bar{B}]^{1/(1-\eta)}, \quad a = \chi B^\eta M^\eta/\bar{B}, \quad r = \alpha y - \delta.$$

Define $g_K = y - c - u - M\tau h^{1/\varphi} - \delta$. The exact equations are

$$\dot{h} = \varphi h(n + \gamma - g_K), \tag{84a}$$

$$\dot{d}/d = g_K - a, \tag{84b}$$

$$\dot{c}/c = n + r - \rho - g_K, \tag{84c}$$

$$\dot{v}/v = r - x/(vB^2) - \eta ad\tau h^{1/\varphi}/B + a - g_K, \tag{84d}$$

$$\dot{\tau} = -(n + \gamma)\tau. \tag{84e}$$

The CES term divided by capital is $[\omega_L h + \omega_X x^\varphi]^{1/\varphi}$. The implicit static solution is smooth in (h, B) near $h = 0$. Since $1/\varphi > 1$, the remaining terms admit a C^1 extension through $h = 0$, sufficient for the stable-manifold argument.

At $h = \tau = 0$, the terminal constants are

$$\begin{aligned} \bar{y} &= z_A(\bar{B}; \sigma), & \bar{u} &= (1 - \alpha)^2 \bar{y}, & \bar{x} &= \bar{B} \bar{u}, \\ \bar{r} &= \alpha \bar{y} - \delta, & \bar{g} &= n + \bar{r} - \rho, & \bar{c} &= \alpha(1 - \alpha) \bar{y} + \rho - n > 0, \\ \bar{v} &= \bar{x} / (\bar{r} \bar{B}^2), & \bar{M} &= \left[\bar{g} \bar{B}^{1-\eta} / \chi \right]^{1/\eta}, & \bar{d} &= \bar{M} / (\eta \bar{g} \bar{v}). \end{aligned} \quad (85)$$

The threshold implies $\bar{g} > n + \gamma > 0$ and $\bar{r} > \rho + \gamma > 0$, so all constants are positive. Substitution verifies the fixed point $(0, \bar{d}, \bar{c}, \bar{v}, 0)$.

The h and τ eigenvalues are $\varphi(n + \gamma - \bar{g}) < 0$ and $-(n + \gamma) < 0$. The consumption-ratio eigenvalue is $\bar{c} > 0$. The remaining block in $(\log d, \log v)$ is

$$\begin{pmatrix} -\bar{g}\mu & -\bar{g}\mu \\ \bar{g}\mu & \bar{r} + \bar{g}\mu \end{pmatrix}, \quad \det = -\bar{g}\mu\bar{r} < 0.$$

The linearized system is triangular in the order $(h, \tau), c, (d, v)$. Choosing c eliminates its growing solution, and choosing v eliminates the growing research solution for a given d . Thus the three-dimensional stable subspace projects invertibly onto (h, d, τ) . The same graph argument constructs the family specified by (45), with a unique nearby convergent choice of C_0, q_0 and positive quantities at every finite date.

Since Y/K , X/K , and C/K converge to positive constants, $g_Y, g_X, g_C, g_K \rightarrow \bar{g}$. Also $s_X \rightarrow 1$ and $1 - s_X$ is asymptotically proportional to $(AN/X)^\varphi$. The wage equation gives

$$g_w \rightarrow \bar{g} - n + \varphi(n + \gamma - \bar{g}) = \gamma + \frac{\bar{r} - \rho - \gamma}{\sigma}.$$

This proves the growth and distribution limits in Proposition 5.

A.6.4 Optimality, transversality, and the scope of existence

In either construction let $\bar{g}$ denote the terminal aggregate growth rate: $n + \gamma$ in the positive labor-share case and $n + \bar{r} - \rho$ in the AI-dominated case. The constructions imply

$$a \rightarrow \bar{g}, \quad M \rightarrow \left[\frac{\bar{g} \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta} > 0, \quad g_q \rightarrow \bar{g}, \quad g_B \rightarrow 0, \quad M/Y \rightarrow 0.$$

Both transversality expressions in Definition 1 have limiting logarithmic growth

$$-\rho + n + g_K - g_C \rightarrow n - \rho < 0, \quad -r + g_q + g_B \rightarrow \bar{g} - \bar{r} = n - \rho < 0. \quad (86)$$

They therefore converge to zero, rather than merely remaining bounded.

Household utility is finite because $\log(C/N)$ grows at most linearly and $\rho > n$. Log utility is concave, the household budget is linear at given prices and distributions, and the Euler and

transversality conditions verify household optimality against admissible budget-feasible alternatives. Final-good optimality follows from concavity and constant returns. The pointwise monopoly optimum is global by Appendix A.2.

For the dynamic developer, feasibility and first-order conditions still require a sufficiency argument. Along the constructed paths, operating profits at any alternative $B \in [B_0, \bar{B})$ grow no faster exponentially than the terminal aggregate scale. The discount rate exceeds that growth rate by $\rho - n > 0$. The candidate's discounted objective is finite, since its research compute converges to a finite constant.

It remains to verify the curvature condition (71) for every date and every alternative AI-efficiency level in $[B_0, \bar{B})$, not just on the selected path. In the positive labor-share construction, $K/(AN)$ stays in a compact neighborhood of $\bar{k} > 0$. Smoothness of the static monopoly solution bounds ε_B uniformly for these states and for AI-efficiency levels near $\bar{B}$. In the AI-dominated construction, h stays near zero, and the same uniform bound follows from the smooth static limit $\varepsilon_B \rightarrow 1/\alpha$. Thus in either case there is a finite constant E with $\varepsilon_B - 2 \leq E$ throughout that state domain. By taking the initial-state neighborhood sufficiently small, B_0 is close enough to $\bar{B}$ that

$$(1 - B_0/\bar{B}) \max\{E, 0\} < B_0/\bar{B}.$$

Because $B \geq B_0$, this inequality implies (71) on the entire alternative-state domain at every date. Together with $2\eta \leq 1$ and the verified transversality condition, Proposition 8 therefore establishes the developer's global optimality.

All quantities remain finite at every finite date, and market clearing and the original initial conditions hold by reconstruction. The paths are therefore infinite-horizon equilibria, not merely solutions of the canonical equations. The result is local only in its set of initial stocks. At $\bar{B} = \bar{B}_c(\sigma)$ the interior-share terminal point reaches the boundary and a stable eigenvalue in the boundary normalization becomes zero; this proof does not apply. For $\sigma < 1$ and $\bar{B} \leq \bar{B}_c(\sigma)$, it does not rule out equilibria with different long-run ratios. Nor does it imply existence at arbitrary initial stocks or in the uncapped limit. $\square$

B Numerical algorithm and equilibrium audit

B.1 Economic intuition and algorithm in words

The numerical problem is to select an equilibrium path, not merely to integrate the laws of motion from four arbitrary initial values. Capital and AI efficiency are predetermined at date zero, whereas consumption and the developer's shadow value can jump. For given K_0 and B_0 , almost every choice of C_0 and q_0 sends the economy away from the terminal equilibrium regime or violates feasibility. The algorithm therefore chooses C_0 and q_0 jointly with the entire path so that the economy starts from the prescribed stocks and approaches the appropriate stable continuation.

The calculation proceeds in six steps:

1. For each value of σ , I use the analytical results in Section 4 to identify the relevant terminal regime and its normalized limiting point.
2. Near that point, I construct a local boundary-value solution. The terminal restrictions eliminate the two unstable directions rather than fixing arbitrary terminal levels for the jump variables.
3. Conditional on every trial value of the dynamic variables, I solve the static monopoly condition for AI production services and recover research expenditure from the developer's first-order condition. Thus the numerical system uses the static choices already derived in Section 3; it does not add behavioral equations.
4. I use continuation to move the local solution gradually to the common prescribed values of K_0 , B_0 , and A_0N_0 . Each successful solution is the initial guess for the next continuation stage. This step selects the two date-zero jump variables without treating the no-AI economy as a mechanical starting point.
5. I lengthen the terminal horizon and tighten the solver tolerances. A valid approximation should leave the date-zero jumps and the entire displayed path essentially unchanged.
6. I admit the path only after separate checks of feasibility, the original equilibrium equations, static optimality, global developer optimality, both transversality conditions, and consistency with the analytical terminal continuation. A converged boundary-value calculation that fails any of these checks is not reported as an equilibrium.

This procedure can be viewed as finding the stable saddle path from both ends. The initial boundary supplies the predetermined stocks, while the terminal boundary rules out explosive components. Collocation chooses the whole path at once and, with it, the initial consumption and shadow value that reconcile the two boundaries. The terminal date is a computational truncation; the infinite-horizon interpretation comes from the analytical continuation and the transversality checks described below.

B.2 Notation in the paper and code

The numerical code retains its existing variable names. To make the implementation auditable, the notation used in the paper maps into the code as follows:

$$\gamma = \text{labor_productivity_growth}, \quad \sigma = \text{sigma_x1}, \quad \psi(B) = \exp(\text{log_psi}). \quad (87)$$

This map changes only the notation in the exposition. It does not redefine a variable or parameter, alter its timing, or modify any equation used by the simulations.

B.3 Boundary-value formulation

A boundary-value problem (BVP) selects the initial consumption and shadow price jointly with the subsequent path. I use the existing collocation solver, `scipy.integrate.solve_bvp`, for the four dynamic equations in the equilibrium definition. At every evaluation, I solve the static monopoly condition and recover research expenditure from its first-order condition. Effective labor is exogenous, $AL = A_0 N_0 e^{(n+\gamma)t}$, and therefore is not an additional unknown path in this dated BVP.

The computational coordinates are

$$\left(\log K, \log \frac{B}{\bar{B} - B}, \log C, \log q \right).$$

They enforce positivity and $0 < B < \bar{B}$ without clipping a computed equilibrium. I subtract the analytical terminal aggregate growth rate times t from each coordinate to remove its linear trend. This is a numerical transformation, not a change in the model's variables. In particular, the code stores $\log \psi(B)$ separately: a displayed ratio $B/\bar{B}$ can round to one even though the remaining frontier gap is strictly positive.

The left boundary fixes only K_0 and B_0 from (56). At the right boundary, the solution is mapped into the regime-specific normalized coordinates of Appendix A.6. Two projection conditions remove the unstable components of the linearized terminal system. Its third stable direction is associated with the exogenous scale. The two stock conditions and two terminal projections thus close a four-dimensional problem; neither C_0 nor q_0 is predetermined. A finite terminal projection approximates the stable continuation, which is why increasing the horizon is a necessary numerical check.

For $\sigma \leq 1$, I use the positive-labor-share terminal formulas. For $\sigma > 1$, the frontier's position relative to $\bar{B}_c$ selects the terminal regime. At $\sigma = 1$, the CES block uses its exact Cobb–Douglas limit. Near one, `log1p` and `expm1` evaluate the CES expression without cancellation, and the frontier comparison is performed in logarithms. I solve the labor-supported terminal static condition directly for $\log[X/(AL)]$, rather than invert a share formula by dividing by $\sigma - 1$. A trial quantity with nonpositive marginal revenue can bracket the monopoly optimum, but is never accepted as that optimum.

B.4 Continuation and numerical accuracy

For each elasticity, I first solve a local BVP near its analytical terminal point. I then move the initial stocks and effective-labor scale continuously to their common prescribed values, retaining the preceding solution as the next guess. The requested design uses 32 continuation stages. There is no continuation through the degenerate no-AI economy, and the uncapped unit-elastic reference supplies initial stocks only.

The initial horizon combines the time needed to reach the terminal scale with eleven decay times of the slowest stable eigenvalue. This gives an initial truncation criterion, not an equilibrium assumption. I extend that horizon twice by 500 years, tightening the collocation tolerance from 2×10^{-6} to 10^{-8} and then 10^{-9} . The final boundary tolerance is 10^{-11} . Initial meshes of 221, 401, and 601 points are refined adaptively by the solver. The final horizons exceed the common 4,000-year display window; the figures contain no extrapolated spline values.

Table 4: Numerical accuracy of the four displayed equilibria

σ	Final horizon	Mesh points	Dynamic residual	Horizon change
0.90	6,857.1	809	8.53×10^{-10}	2.44×10^{-8}
1.00	6,731.9	607	8.88×10^{-10}	5.73×10^{-9}
1.10	6,647.9	831	9.37×10^{-10}	1.68×10^{-8}
1.50	4,919.4	1,255	1.06×10^{-9}	2.73×10^{-8}

Dynamic residuals reconstruct the four original equations using five-point differences of spline values at 1,001 dates, with steps of 0.003 and 0.001 years; the table reports the larger maximum. Horizon change is the largest change in the detrended logarithmic coordinates over years 0–4,000 between the last two solves. Values are rounded upward.

The maximum research and static monopoly first-order residuals are below 2.3×10^{-13} . Initial jump choices change by less than 2.4×10^{-11} in logs in the final horizon comparison. The normalized terminal coordinates differ from their analytical limits by less than 3.2×10^{-6} . The acceptance thresholds are deliberately looser than these observed errors: 10^{-6} for independently reconstructed dynamic residuals, 10^{-9} for static first-order conditions, 2×10^{-5} for horizon sensitivity, and 10^{-4} for terminal-coordinate gaps. Tightening the solver and extending its horizon provide the evidence of stability; the thresholds do not replace any economic condition.

B.5 Optimality and the infinite-horizon continuation

The equation audit also checks feasibility, the monopoly second-order condition, and that numerical safeguards used for off-path Newton trials do not bind on the solution. Both transversality expressions converge to zero on each selected continuation: their asymptotic log growth rate is $n - \rho = -0.037$. This analytical rate, rather than a small terminal value alone, establishes the transversality check.

For $\sigma = 0.90, 1, 1.10$, the developer’s operating profit is concave in research depth over the counterfactual AI-efficiency domain checked at each date. The final audit uses 161 dates and 161 AI-efficiency points per date, with scalar minimization between grid points. For $\sigma = 1.50$, this

stronger curvature test fails at low counterfactual AI-efficiency levels; checking concavity only along the realized path would miss that failure.

I instead apply Proposition 9. At each date, I minimize its tangent gap, divided by the candidate’s Y , over counterfactual AI-efficiency levels $B \geq B_0$, refining every local minimum found on the grid. An analytical upper bound on the Hamiltonian’s derivative covers the unbounded research-depth domain beyond the numerical grid. The check is repeated with 81 dates and 101 AI-efficiency points, then with 321 dates and 241 points. The smallest normalized gap is -2.6×10^{-14} , consistent with floating-point error at the tangent point. The numerical gap tolerance is 10^{-10} ; the economic support inequality remains exact.

For the infinite-horizon continuation, I use (75) with $B_* = 0.75\bar{B}$. Its limiting margin divided by output is approximately $0.09781 > 0$; the computed terminal margin differs from that limit by less than 4×10^{-13} . Thus the analytical continuation satisfies the sufficient support condition eventually, while the dated audit checks the computed transition. These are numerically verified equilibrium approximations supported by analytical terminal and optimality results, not interval-arithmetic existence certificates for the common initial stocks. Nor do they establish existence for every initial condition or every parameter vector.

B.6 Reproduction and export safeguards

The implementation preserves the earlier solver. The entry point is `scripts/simulate_rewrite_finite_frontier.py`; the final audit is `scripts/audit_rewrite_equilibria.py`. The reproducible sequence is: solve each of the four elasticities with `--sigma`; run `--verify-long-horizon`; run `audit_rewrite_hamiltonian_support.py` for $\sigma = 1.50$ at both grid resolutions; run the final audit; export with `--export-horizon 4000`; and render with `plot_rewrite_equilibria.py`. All code, audit files, plot-ready data, and figures used in these simulations are available in the paper’s [public GitHub repository](#). The repository’s [replication guide](#) provides a one-command workflow, installation instructions, an output map, and the criteria used to interpret the diagnostics. The execution record `audit/rewrite_equilibrium_simulations.md` reports the commands and test results for the version used in the paper.

The exported data and figures require all four scenarios to pass admission. Checkpoint hashes link each audit to its solved spline; a second manifest links the plotted CSV to those checkpoints. Failed or stale candidates cannot silently enter a partial comparison. The data retain the original model variables and use only the normalizations stated in the figure captions.

C Conditional and excluded limits

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